PROJECT REPORT ON

CONSTRUCTION HELPER

Submitted to the university of Calicut in partial fulfillment of the requirement for the award of degree of

BACHELOR OF COMPUTER APPLICATION OF UNIVERSITY OF CALICUT

2021-2024

Done By

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Under the guidance of

Mrs. INDHU

DEPARTMENT OF COMPUTER APPLICATION MAR OSTHATHEOS COLLEGE, PERUMPILAVU THRISSUR DIST, KERALA

CERTIFICATE

This is to certify that the project entitled **"CONSTRUCTION HELPER "** is an authentic record of project work done by

ASHIK E.A

Under the guidance of

Mrs.INDHU

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF BACHELOR OF COMPUTER APPLICATION OF UNIVERSITY OF CALICUT

Head of Department **Faculty Guide**

Submitted for viva-voice Examination held on .....

Internal Examinar **External Examinar**

DECLARATION

We hereby declare that this project work entitled **"CONSTRUCTION HELPER"** submitted to the University of Calicut in partial fulfillment of the requirements for the award of degree in bachelor of computer application is a record of original work done by us, under the guidance of **Mrs. INDHU** Assistant professor, Mar Osthatheos College Perumpilavu.

ASHIK E.A

Place : Perumpilavu

Date :

ACKNOWLEDGEMENT

We wish to take this opportunity to express our sincere thanks to all Who contributed to this project and helped us at every turn.

Firstly and foremost we would like to convey our sincere thanks to **Prof. BABU C. P** Principal. Mar Osthatheos College, Perumpilavu for his help. We would like to express our sincere thanks to **Mrs. INDHU** Assistant professor of computer science for her assistance and guidance throughout the project. We are indebted to her for the valuable discussions and helpful counseling.

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Above all we the Almighty for all he has done for us .....

ASHIK E.A

SYNOPSIS

The construction helper mobile application aims to bring all the needs for constructing a building in one single mobile application. The mobile application uses android -java at the front-end and PHP-MYSQL at the back end. The application provides an interface to the users in which the user can enter input parameters like number of rooms, carpet area, total number of space for the house in square feet etc and they will get the plan as the result back.

There is a human resource management section so that the user can search nearby resource persons such as plumbers, electricians, painters etc in the order of their location. They can hire them if needed by clicking the hire button.They can also provide reviews and ratings to the services provided by the workers. The system provides a searching facility to the customers to choose the particular home service product from the nearest store in a short time (customized).

Users / Actors Of System

1. Admin
2. User
3. Builder
4. Merchant

Admin

The module "Admin" typically refers to a crucial component within a project construction helper system. Its primary function is to provide administrative functionalities necessary for managing various aspects of the project. Here's a comprehensive description of what the "Admin" module might entail:

- User Management: The Admin module facilitates the management of user accounts within the system.
- Workflow Management: It enables administrators to define and customize project workflows according to the specific requirements of the construction project.
- Maintenance and Support: The Admin module may also include functionalities for system maintenance, such as backup and recovery procedures, software updates, and providing technical support to users to ensure smooth operation of the platform

User

The User Module focuses on the individuals or entities who require construction services. These users could be homeowners, businesses, or organizations seeking to initiate construction projects.

● Features of the User Module might include:
● Project Creation: Users can create new construction projects, specifying details such as project type, location, budget, and timeline.
● Search and Browse: Users can search for builders, contractors, and merchants based on specific criteria such as expertise, location, ratings, and reviews.
● Communication: Users can communicate with builders, contractors, and merchants to discuss project requirements, obtain quotes, and clarify details. Feedback and Reviews: Users can provide feedback and reviews based on their experience with builders, contractors, and merchants, helping other users make informed decisions.

Builder

The Builder Module caters to professionals or companies involved in construction activities, including contractors, construction firms,architects, and engineers.

Features of the Builder Module might include:

● Profile Creation: Builders can create profiles showcasing their expertise, qualifications, past projects, and client testimonials.
● Project Management: Builders can manage ongoing projects, including scheduling, resource allocation, task assignment, and progress tracking.

● Communication: Builders can communicate with users to understand project requirements, provide updates, and address concerns.

● Document Management: Builders can manage project documents, contracts, permits, and regulatory compliance.
● Quality Control: Builders can ensure adherence to quality standards, safety regulations, and industry best practices throughout the construction process..

Merchant

The Merchant Module focuses on suppliers, vendors, and manufacturers of construction materials, equipment, and supplies.

Features of the Merchant Module might include:

● Product Catalog: Merchants can create and manage a catalog of construction materials, equipment, and supplies available for purchase or rental.
● Order Management: Merchants can process orders, track inventory levels, and manage deliveries to construction sites.
● Quotation System: Merchants can provide quotes and pricing information to users and builders based on their requirements and quantities.
● Invoice Generation: Merchants can generate invoices, manage billing cycles, and track payments for orders placed by users and builders.
● Supplier Collaboration: Merchants can collaborate with other suppliers or manufacturers to fulfill large orders or provide specialized products.
● Customer Support: Merchants can provide customer support, address inquiries,and resolve issues related to orders, deliveries, and product inquiries..

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INTRODUCTION

1. INTRODUCTION

The construction helper mobile application aims to bring all the needs for constructing a building in one single mobile application. The mobile application uses android -java at the front-end and PHP-MYSQL at the back end. The application provides an interface to the users in which the user can enter input parameters like number of rooms, carpet area, total number of space for the house in square feet etc and they will get the plan as the result back.

There is a human resource management section so that the user can search nearby resource persons such as plumbers, electricians, painters etc in the order of their location. They can hire them if needed by clicking the hire button .They can also provide reviews and ratings to the services provided by the workers. The system provides a searching facility to the customers to choose the particular home service product from the nearest store in a short time (customized).

1.1 PURPOSE

The purpose of a construction helper project typically involves assisting in various tasks related to construction projects, such as carrying materials, construct a new buildings ,to connect merchants and products

The ultimate goal is to contribute to the smooth execution and completion of construction projects by providing assistance where needed Their contributions help enhance productivity, improve job site safety, budget friendly and ultimately contribute to the timely completion of construction projects. Additionally, construction helper projects provide valuable learning opportunities for individuals interested in pursuing careers in construction by allowing them to gain hands-on experience and develop essential skills in the field

1.2 OBJECTIVE

Reduce Time to Construct Buildings: The primary aim here is to optimize the construction process to decrease the time it takes to complete building projects. This can involve various strategies such as implementing efficient project management techniques, utilizing advanced construction technologies Provide a Platform for Customers to Compare Construction Companies with Feedback: This objective focuses on empowering customers by providing them with a platform where they can access and compare different construction companies based on feedback and reviews.

These objectives aim to streamline the construction process and enhance customer satisfaction through transparency and informed decision-making.

1.3 MODULES

There are 4 Modules

- ADMIN
- USER
- BUILDER
- MERCHANT

ADMIN MODULE

The module "Admin" typically refers to a crucial component within a project construction helper system. Its primary function is to provide administrative functionalities necessary for managing various aspects of the project. Here's a comprehensive description of what the "Admin" module might entail.

USER MODULE

The User Module focuses on the individuals or entities who require construction services. These users could be homeowners, businesses, or organizations seeking to initiate construction projects.

BUILDER MODULE

The Builder Module caters to professionals or companies involved in construction activities, including contractors, construction firms ,architects, and engineers. Features of the Builder Module might include:
Profile Creation, Project Management ,Communication , Document Management Quality Control.

MERCHANT MODULE

The Merchant Module focuses on suppliers, vendors, and manufacturers of construction materials, equipment, and supplies. Features of the Merchant Module might include: Product Catalog: Merchants can create and manage a catalog of construction materials, equipment, and supplies available for purchase or rental.

SYSTEM STUDY AND ANALYSIS

2.SYSTEM STUDY AND ANALYSIS

2.1 EXISTING SYSTEM

The existing system might be characterized by traditional methods of managing construction projects, often involving manual processes, fragmented communication, and limited access to information. Here are some key points. Fragmented Communication: Communication between stakeholders such as users, builders, and merchants may occur through phone calls, emails, or in-person meetings, leading to delays, misunderstandings, and inefficiencies. Limited Access to Information: Users may face challenges in finding reliable builders, contractors, and merchants, while builders and merchants may struggle to reach potential clients due to limited visibility..

2.2 PROPOSED SYSTEM

The proposed system aims to address the shortcomings of the existing system by leveraging modern technology and digital tools to streamline project management, enhance communication, and improve accessibility. Here are some key features of the proposed system: Integrated Platform: The proposed system will provide an integrated platform where users, builders, and merchants can connect and collaborate throughout the construction project lifecycle. Communication Channels: The system will offer various communication channels such as messaging, video conferencing, and document sharing to facilitate seamless communication between stakeholders

2.3 FEASIBILITY STUDY

A feasibility study for a construction helper project is essential to determine whether the proposed system is viable, practical, and beneficial. It involves assessing various aspects of the project, including technical, economic, operational, legal, and scheduling considerations. Here's a breakdown of the feasibility study for a construction helper project::

2.4 TECHNICAL FEASIBILITY

Evaluate the technical requirements of developing and implementing the construction helper platform, including hardware, software, and networking infrastructure. Assess the availability of technology platforms, development tools, and expertise required to build and maintain the system. Consider compatibility with existing systems, databases, and software used by stakeholders in the construction industry.

2.5 ECONOMIC FEASIBILITY

Estimate the initial investment required for developing the construction helper platform, including software development costs, hardware expenses, licensing fees, and staffing. Conduct a cost-benefit analysis to determine the potential return on investment (ROI) and the financial viability of the project. Consider factors such as projected revenue streams, subscription models, advertising opportunities, and potential cost savings for users, builders, and merchants.

2.6 OPERATION FEASIBILITY

Estimate the initial investment required for developing the construction helper platform, including software development costs, hardware expenses, licensing fees, and staffing. Conduct a cost-benefit analysis to determine the potential return on investment (ROI) and the financial viability of the project. Consider factors such as projected revenue streams, subscription models, advertising opportunities, and potential cost savings for users, builders, and merchants.

2.7 SOFTWARE REQUIREMENT SPECIFICATION

Functional Requirements

Input Requirements

The system shall prompt the user to enter the first number (a).

The system shall prompt the user to enter the second number (b).

The system shall prompt the user to choose an operation (addition, subtraction, multiplication, or division).

Processing Requirements

The system shall perform addition when the user selects option 1.

The system shall perform subtraction when the user selects option 2.

The system shall perform multiplication when the user selects option 3.

The system shall perform division when the user selects option 4.

Output Requirements

The system shall display the result of the selected operation.

The system shall prompt the user if they want to continue using the calculator after displaying the result

Non-Functional Requirements

Usability

The user interface shall be intuitive and easy to navigate.

The system shall provide clear instructions for user inputs and choices.

The system shall handle user errors gracefully and provide helpful error messages.

Performance

The system shall respond to user inputs without noticeable delay.

The system shall accurately perform arithmetic operations within a reasonable time frame.

Reliability

The system shall produce accurate results for all valid inputs.

The system shall handle unexpected errors or edge cases effectively.

Portability

The system shall be compatible with various Unix-like operating systems, including Linux and macOS.

The system shall be implemented using portable scripting languages and tools.

SYSTEM REQUIREMENTS

3. SYSTEM REQUIREMENTS

System configuration mainly refers to the specification of a given computer system, from its hardware components to the software and various processes that are run within that system. It refers to what types and models of devices are installed and what specific software is being used to run the various parts of the computer system. By extension, system configuration also refers to the specific operating system settings that have been set by default automatically or manually by a given program or the user.

3.1 Hardware Requirements

The hardware is the place where all the information and data are store permanently. So hardware must be reliable and cost effective. The hardware must suit all the application development. It is fast enough to complete and do all the jobs and executions.

Server: .Processor Intel Pentium .

RAM: 12GB.

Hard Disk: 1TB CPU Speed: 2.60GHz.

Monitor Client: VGA Color.

Processor i3 7th GEN.

Hard Disk 1TB CPU.

Speed 160GHz.

Monitor VGA Color

3.2 Software Requirements

The software specification means the operating system and all other application or tools used for the development of the proposed system. It includes the operating system, and the software which we are going to use.

Web Server:

Apache Server front end tool :android-java

platform

: android studio

- back end tool :php-mysql, python
- operating system :windows 10
- server :c panel

Client:BrowserEdge/Chrome/Firefox

SOFTWARE CONFIGURATION

4. SOFTWARE CONFIGURATION

4.1ABOUT WINDOWS

Windows 10

Windows 10 is a Microsoft operating system for personal computers, tablets, embedded devices and internet of things devices.

Microsoft released Windows 10 in July 2015 as a follow-up to Windows

8. The company has said it will update Windows 10 continuously, rather than release a new, full-fledged operating system as a successor. The tools and technologies used during the development and implementation of our project include:

Anyone adopting Windows 10 can upgrade legacy machines directly from Windows 7 or Windows 8 to Windows 10 without re-imaging or performing intrusive and time-consuming system wipes and upgrade procedures. To upgrade from a previous version of Windows 10, IT or users run the Windows 10 OS installer, which transfers any applications and software on the previous OS, as well as settings and preferences over to Windows 10. Organizations and users can pick and choose how they will patch and update Windows 10. IT or users can access a Windows 10 upgrade through the Windows Update Assistant to manually begin an upgrade or wait for Windows Update to offer an upgrade when it is set to run.

4.2 ABOUT FRONT END

Android - java

Java holds a significant space in the realm of Android development. While Kotlin has emerged as a popular alternative, Java's roots run deep, serving as the foundation for many existing apps. It empowers developers to construct the core functionalities of an Android app, encompassing aspects like user interaction, data management, and seamless integration with the Android system itself. While the realm of UI development has gravitated towards XML layouts, Java provides the essential framework that breathes life into the user experience, making the app interactive and responsive. In essence, Java serves as the backbone, enabling developers to craft the logic and functionality that powers the user experience on countless Android devices.

4.3ABOUT BACK END

PHP

PHP It stands for PHP: Hypertext Preprocessor but originally stood for Personal Home Page. Is a server side scripting language that is designed for web development, as well as used for general purpose language. It was created in 1994 by Rasmus Lerdorf, in the present time the reference execution of PHP is produced by the PHP group.

PhpMyAdmin It is an open source tool and also, it is free written in PHP, XHTML, CSS, and JavaScript planned to manage the administration of MySQL by using a web. It is able to perform various missions like creating, modifying databases, tables, fields, executing SQL statements or managing and supervising users.

PhpMyAdmin is being translated into 72 languages in order to make the usage easy to a wide domain of people and it supports both LTR and RTL languages. Following is some features of the phpMyAdmin,

- ❖ It is web interface
- ❖ It administers multiple servers
- ❖ It is able to create PDF graphics of the database layout
- ❖ Importing data from SQL and CSV
- ❖ Export data to different formats such as SQL, PDF, CSV, XML and others
- ❖ It works with various Operating Systems.

Syntax

A PHP script starts with <?php and ends with?>:

<?php

/ PHP code goes here

?>

MySQL

MySQL is a database management system that allows you to manage relational databases. It is open source software backed by Oracle. It means you can use MySQL without paying a time. Also, if you want, you can change its source code to suit your needs.Even though MySQL is open source software, you can buy a commercial license version from Oracle to get premium support services.MySQL is pretty easy to master in comparison with other database software like Oracle Database, or Microsoft SQL Server.MySQL can run on various platforms UNIX, Linux, Windows, etc.

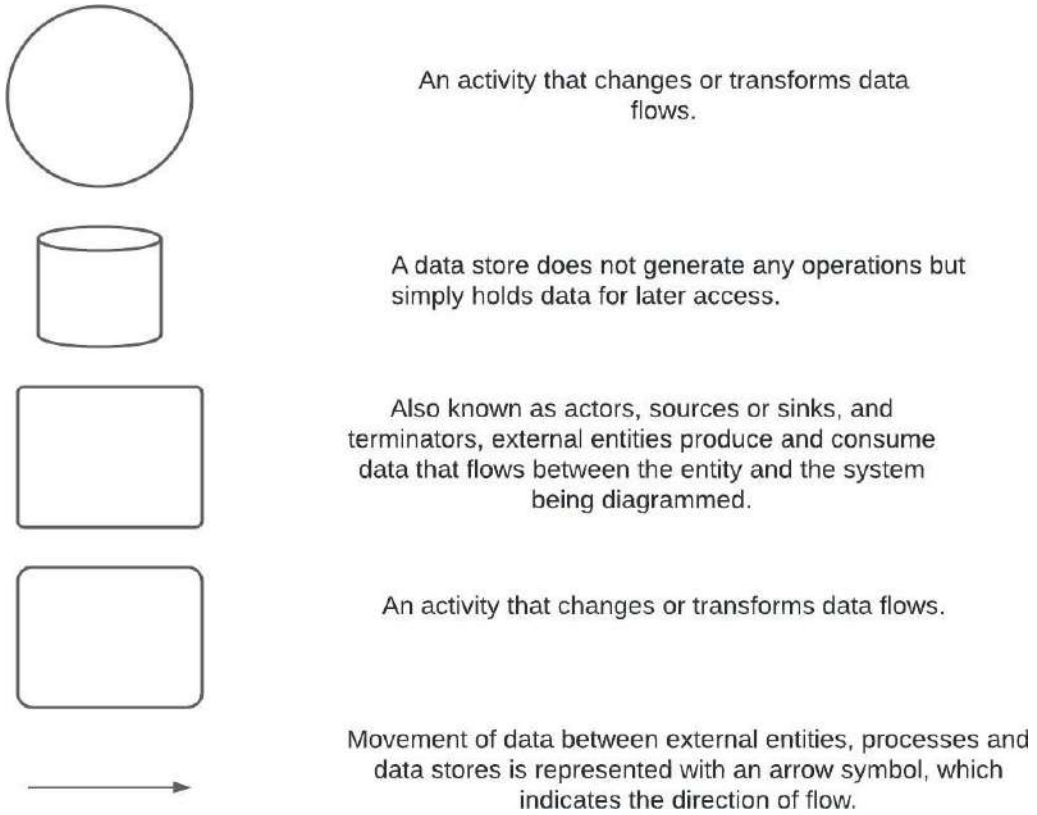
You can install it on a server or even on a desktop. Besides, MySQL is reliable, scalable, and fast. SQL is the most popular language for adding, accessing and managing content in a database. It is most noted for its quick processing, proven reliability, ease and flexibility of use. MySQL is an essential part of almost every open source PHP application. Good examples for PHP & MySQL-based scripts are WordPress, Joomla!, and Drupal.

DATABASE DESIGN

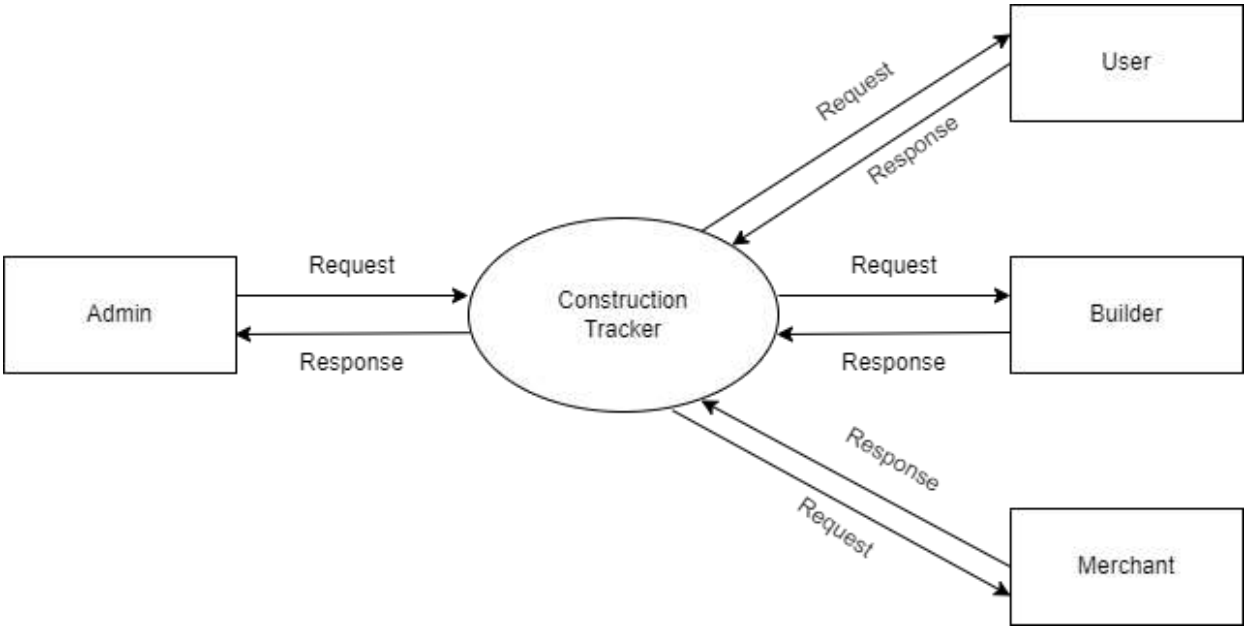
5 DATABASE DESIGN

5.1 DATA FLOW DIAGRAM (DFD)

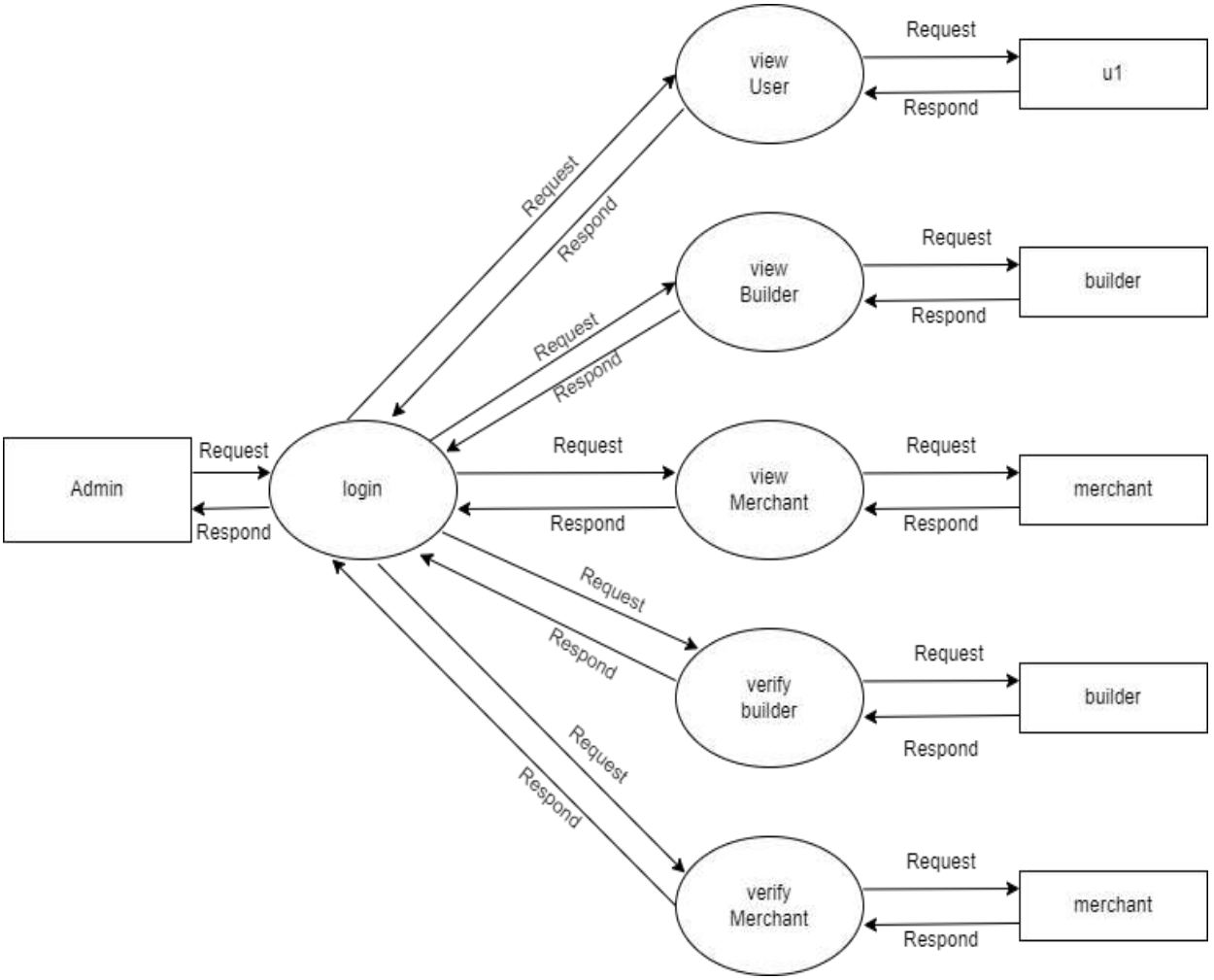
A data flow diagram (DFD) maps out the flow of information for any process or system. It uses defined symbols like rectangles, circles and arrows, plus short text labels, to show data inputs, outputs, storage points and the routes between each destination. Data flowcharts can range from simple, even hand-drawn process overviews, to in-depth, multi-level DFDs that dig progressively deeper into how the data is handled. They can be used to analyze an existing system or model a new one. Like all the best diagrams and charts, a DFD can often visually "say" things that would be hard to explain in words, and they work for both technical and nontechnical audiences, from developer to CEO. That's why DFDs remain so popular after all these years.



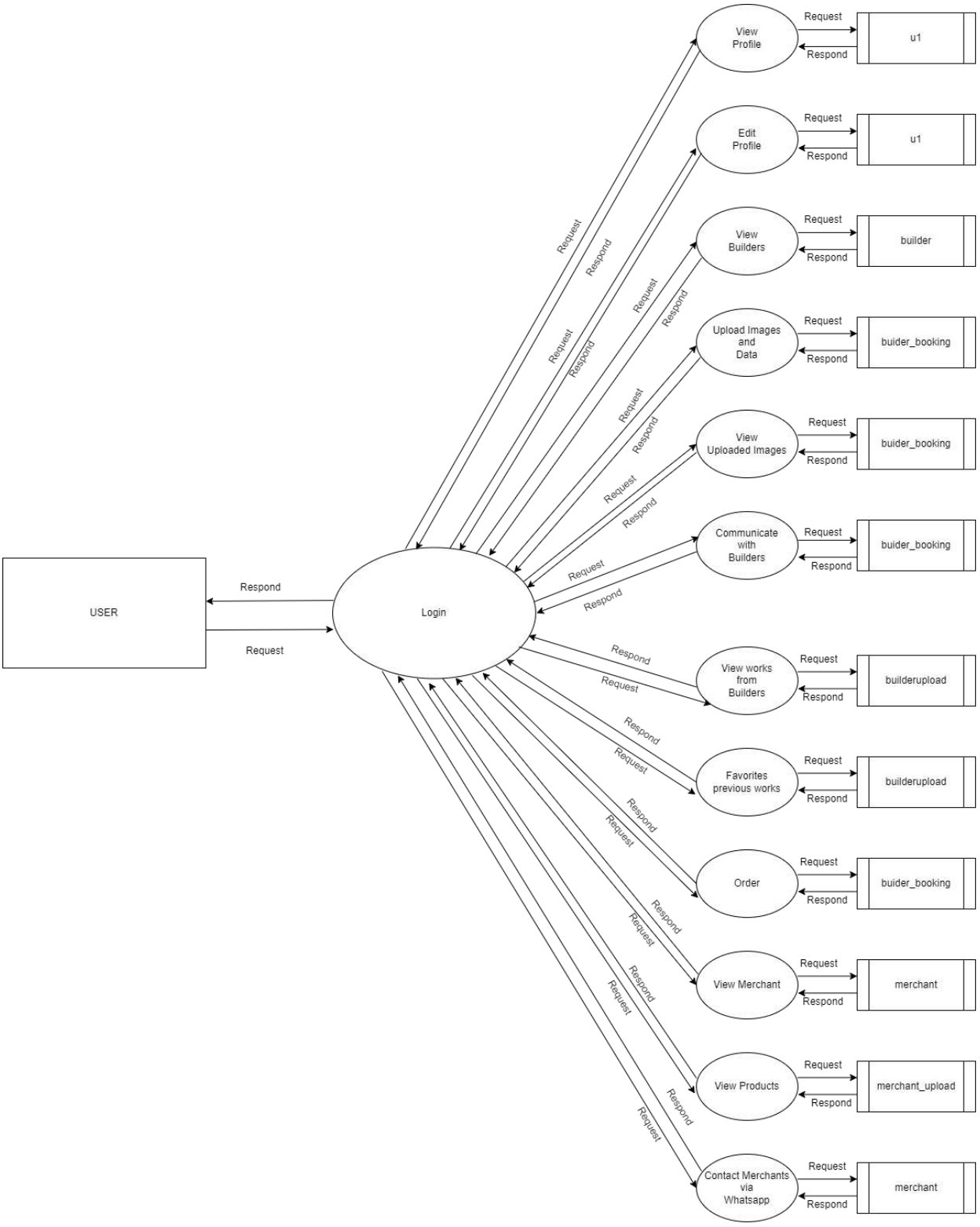
LEVEL 0 DFD


User
Request
Response
Admin
Request
Response
Construction
Tracker
Request
Response
Builder
Response
Request
Merchant

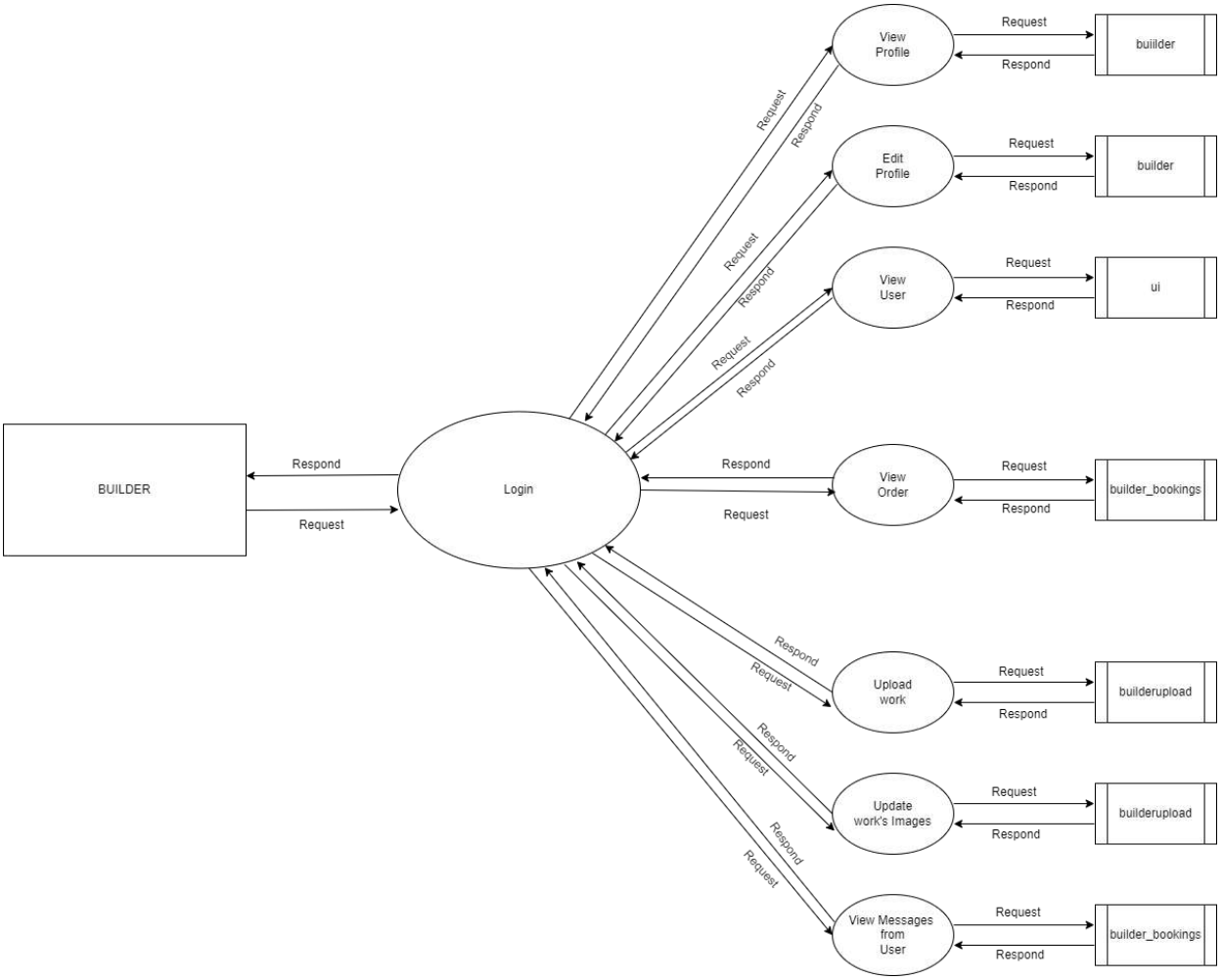
LEVEL 1 DFD ADMIN



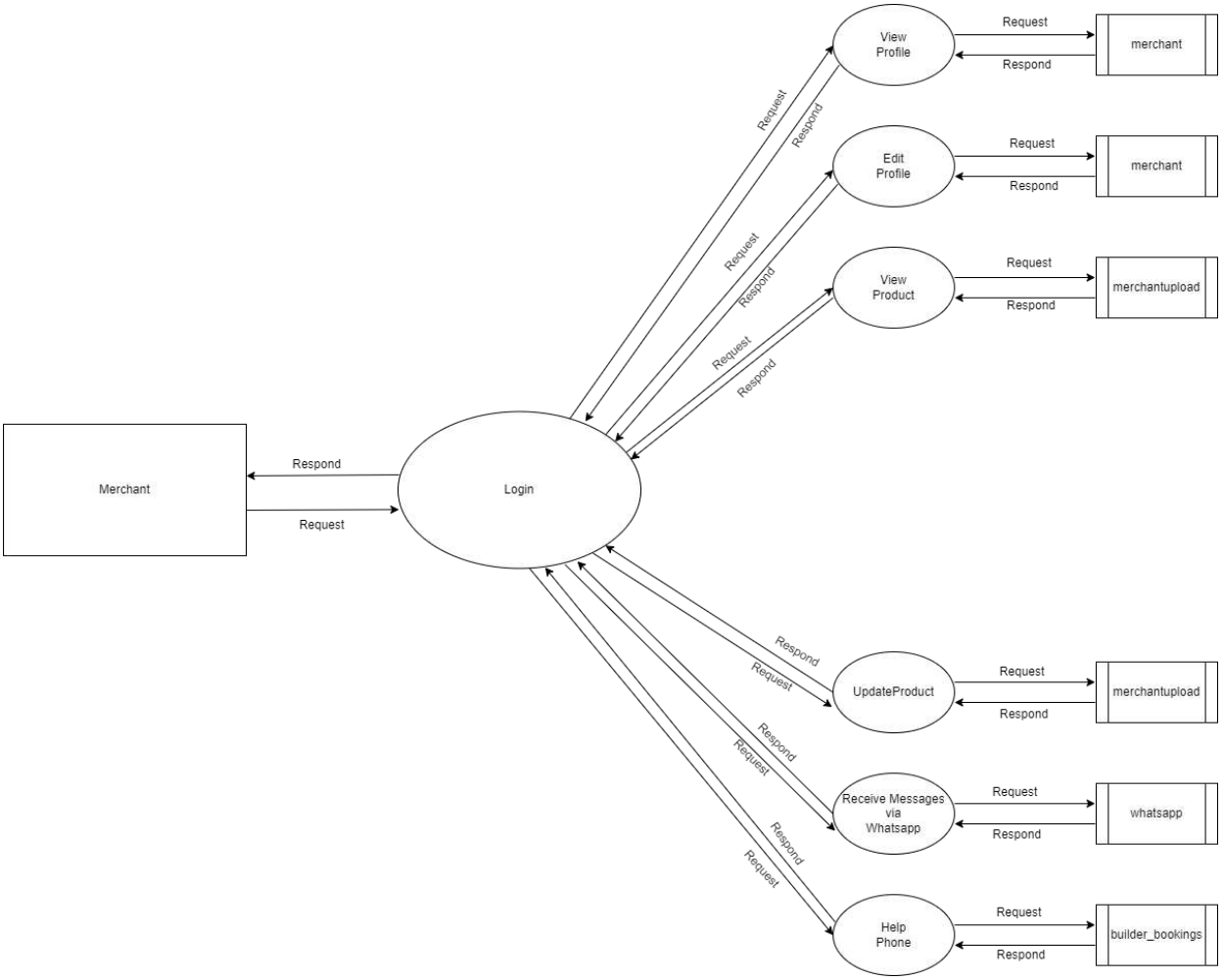
LEVEL 2 DFD USER



LEVEL 3 DFD BUILDER



LEVEL 4 DFD MERCHANT



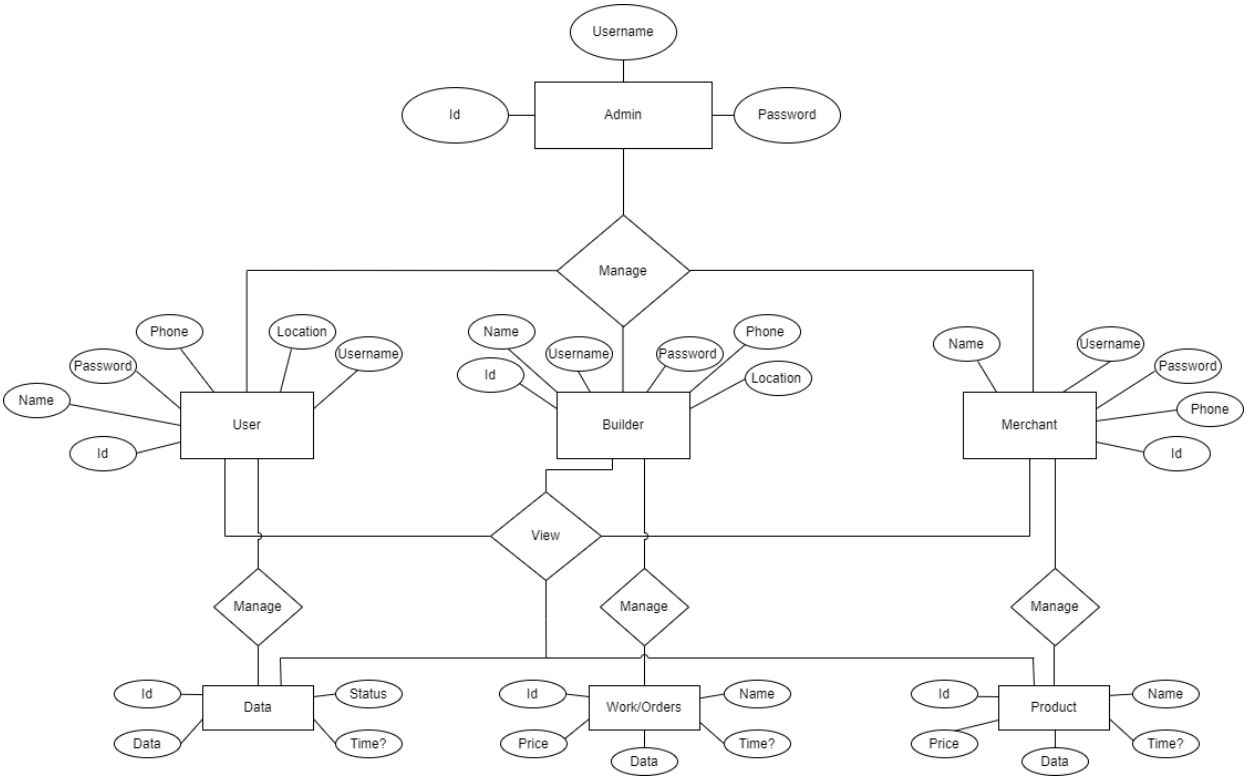
5.2 E R DIAGRAM

An **Entity–relationship model (ER model)** describes the structure of a database with the help of a diagram, which is known as **Entity Relationship Diagram (ER Diagram)**. An ER model is a design or blueprint of a database that can later be implemented as a database. The main components of the E-R model are: entity set and relationship set.

An ER diagram shows the relationship among entity sets. An entity set is a group of similar entities and these entities can have attributes. In terms of DBMS, an entity is a table or attribute of a table in a database, so by showing relationships among tables and their attributes, the ER diagram

shows the complete logical structure of a database. Let's have a look at a simple ER diagram to understand this concept.

Entity–relationship modeling was developed for database and design by Peter Chen and published in a 1976 paper, with variants of the idea existing previously. Some ER models show super and subtype entities connected by generalization-specialization relationships, and an ER model can be used also in the specification of domain-specific ontologies



NORMALISATION

Normalization is the process of reorganizing data in a database so that it meets two basic requirements:

- There is no redundancy of data, all data is stored in only one place.
- Data dependencies are logical, all related data items are stored together

Database Normal Forms

- 1NF (First Normal Form)
- 2NF (Second Normal Form)
- 3NF (Third Normal Form)
- BCNF (Boyce-Codd Normal Form)
- 4NF (Fourth Normal Form)
- 5NF (Fifth Normal Form

1NF (First Normal Form) Rules

An entity is in the first normal form if it contains no repeating groups. In relational terms, a table is in the first normal form if it contains no repeating columns. Repeating columns makes your data less flexible, waste disk space, and makes it more difficult to search for data.

➢ Each table cell should contain a single value.

➢ Each record needs to be unique

2NF (Second Normal Form) Rules

Tables in 2NF must be in 1NF and not have any partial dependency (e.g. every non-prime attribute must be dependent on the table's primary key).

➢ Rule 1- Be in 1NF

➢ Rule 2- Single Column Primary Key

Foreign Key references the primary key of another Table. It helps connect your Table

3NF (Third Normal Form) Rules

Tables in 3NF must be in 2NF and have no transitive functional dependencies on the primary key.

- Rule 1- Be in 2NF
- Rule 2- Has no transitive functional dependencies

- To move our 2NF table into 3NF, we again need to again divide our table.

BCNF (Boyce-Codd Normal Form)

A higher version of the 3NF, the Boyce-Codd Normal Form is used to address the anomalies which might result if one more than one candidate key exists. Also known as 3.5 Normal Form, the BCNF must be in 3NF and in all functional dependencies ($X \rightarrow Y$), X should be a super key. Even when a database is in 3rd Normal Form, still there would be anomalies resulting if it has more than one Candidate Key. Sometimes BCNF is also referred to as 3.5 Normal Form.

All these above mentioned normal forms are applied in our tables. Primary keys are used in all tables and Reference key also applied in the tables. This includes creating tables and establishing relationships between those tables according to rules designed both to protect the data and to make the database more flexible by eliminating redundancy ana inconsistent dependency.

TABLE DESIGN

BUILDER

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
name	varchar(100)	Not null
place	varchar(100)	Not null
phone_number	varchar(100)	Not null
email	varchar(100)	Not null
username	varchar(100)	Not null
password	varchar(100)	Not null
district	varchar(100)	Not null
status	varchar(100)	Not null
design1	varchar(100)	Not null
design2	varchar(100)	Not null
document	varchar(100)	Not null

MERCHANT

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
name	varchar(100)	Not null
place	varchar(100)	Not null
phone_number	varchar(100)	Not null
email	varchar(100)	Not null
username	varchar(100)	Not null
password	varchar(100)	Not null
district	varchar(100)	Not null
category	varchar(100)	Not null
status	varchar(100)	Not null
document	varchar(100)	Not null

U1

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
name	varchar(100)	Not null
place	varchar(100)	Not null
phone_number	varchar(100)	Not null
email	varchar(100)	Not null
age	varchar(100)	Not null
username	varchar(100)	Not null
password	varchar(100)	Not null

Admin

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
username	varchar(100)	Not null
password	varchar(100)	Not null

BUILDER UPLOAD

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
name	varchar(100)	Not null
place	varchar(100)	Not null
phone_number	varchar(100)	Not null
email	varchar(100)	Not null
image	varchar(100)	Not null

BUILDER BOOKING

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
budget	varchar(100)	Not null
squarefeet	varchar(100)	Not null
placeaddress	varchar(100)	Not null
builderid	varchar(100)	Not null
buildername	varchar(100)	Not null
userid	varchar(100)	Not null
username	varchar(100)	Not null
userphn	varchar(100)	Not null
image	varchar(100)	Not null

UPLOAD

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
placeaddress	varchar(100)	Not null
squarefeet	varchar(100)	Not null
budget	varchar(100)	Not null

USER COMPLAINT

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
section	varchar(100)	Not null
phone	varchar(100)	Not null
complaint	varchar(100)	Not null

USER WATCH LIST

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
builderid	varchar(100)	Not null
name	varchar(100)	Not null
place	varchar(100)	Not null
phone_number	varchar(100)	Not null
email	varchar(100)	Not null
district	varchar(100)	Not null
design1	varchar(100)	Not null
design2	varchar(100)	Not null
userid	varchar(100)	Not null

MERCHANT UPLOAD

FIELD NAME	TYPE	CONSTRAINTS
id	varchar(100)	Primary key
name	varchar(100)	Not null
phone	varchar(100)	Not null
district	varchar(100)	Not null
category	varchar(100)	Not null
p1	varchar(100)	Not null
p1c1	varchar(100)	Not null
p1c2	varchar(100)	Not null
p1c3	varchar(100)	Not null
p2	varchar(100)	Not null
p2c1	varchar(100)	Not null
p2c2	varchar(100)	Not null
p2c3	varchar(100)	Not null
p3	varchar(100)	Not null
p3c1	varchar(100)	Not null
p3c2	varchar(100)	Not null
p3c3	varchar(100)	Not null

INPUT DESIGN

The input design is the link between the information system and the user. It comprises the developing specification and procedures for data preparation and those steps are necessary to put transaction data into a usable form for processing can be achieved by inspecting the computer to read data from a written or printed document or it can occur by having people keying the data directly into the system. The design of input focuses on controlling the amount of input required, controlling the errors, avoiding delay, avoiding extra steps and keeping the process simple. The input is designed in such a way so that it provides security and ease of use with retaining privacy.

OUTPUT DESIGN

A quality output is one, which meets the requirements of the end user and presents the information clearly. In any system results of processing are communicated to the users and to other systems through outputs. In output design it is determined how the information is to be displaced for immediate need and also the hard copy output

SYSTEM TESTING AND IMPLEMENTATION

6. SYSTEM TESTING AND IMPLEMENTATION

6.1 TYPES OF TESTING

System testing is a critical element of a software quality assurance and represents the ultimate reviews of specification, design and coding. Testing present an interesting anomaly for the software. Testing is vital to the success of the system. Error can be injected at any stage during development. System testing makes logical assumptions that if all the parts of the system is correct the goal will be successfully achieved. During testing the program to be tested is executed with set of data and the output of the program for test is evaluated to determine if the program is performing as expected. A series of testing are performed for the proposed system before the system is ready for user acceptance testing. The testing steps are:

- Unit testing
- Integration testing
- System testing
- Acceptance testing

Unit Testing

Unit testing focuses verification effort on the smallest unit of the software design, the modules this is known as module testing.Since the proposed system has four modules, the testing is individually performed on the each module. This testing step each module is found to be working satisfactory as regards to the expected output from the module.

Integration Testing

Data can be test across an interface, one module can have adverse effect on another, sub function when combined may not produce the desired function Integration testing is a systematic technique for constructing the program structure while at the same time conducting test to uncover errors associated within the interface. Integration testing is successfully completed.

System Testing

System testing or end to end, test a completely integrated system to verify that it meets its requirement. For example, a system test might involve a testing login interface, then creating and editing an entry, plus sending or printing result, followed by summery processing or deletion entities. After the entities modules are completely coded they are integrated and test final result.

Acceptance Testing

Acceptance is used to conduct of a product, services or system as apart of quantity management system. Acceptance testing is a common type of non-functional software testing ,used mainly software development and software maintenance projects, this type of testing focuses the pre release of the system to be supported and/or to become part of the production environment functional testing is limited to those test are require to verify the non-functional aspects. The software testing should ensure that portability of the system.

Black box testing is a software testing technique that focuses on the analysis of software functionality, versus internal system mechanisms. White Box Testing usually performed by the developer or someone who has knowledge on the coding language used in the given application

6.2 IMPLEMENTATION

The implementation process (putting the network design into practice) is probably the most critical stage of the project, as it requires a huge commitment in terms of manpower and financial resources, and can be quite disruptive of the day to day operation of the organisation. Implementation is the process that actually yields the lowest-level system elements in the system hierarchy (system breakdown structure). System elements are made, bought, or reused. Production involves the hardware fabrication processes of forming, removing, joining, and finishing, the software realization processes of coding and testing, or the operational procedures development processes for operators' roles. If implementation involves a production process, a manufacturing system which uses the established technical and management processes may be required. The purpose of the implementation process is to design and create (or fabricate) a system element conforming to that element's design properties and/or requirements. The element is constructed employing appropriate technologies and industry practices. This process bridges the system definition processes and the integration process. To implement a system successfully, many interrelated tasks need to be carried out in an appropriate sequence. Utilising a well-proven implementation methodology and enlisting professional advice can help but often it is the number of tasks, poor planning and inadequate resourcing that causes problems with an implementation project, rather than any of the tasks being particularly difficult. Similarly with the cultural issues it is often the lack of adequate consultation and two-way communication that inhibits achievement of the desired results.

SYSTEM SECURITY

7. SYSTEM SECURITY

The purpose of Security Tests is to identify all possible loopholes and weaknesses of the software system which might result in a loss of information, revenue, repute at the hands of the employees or outsiders of the Organization.

7.1 DATABASE SECURITY

Database security refers to the range of tools, controls, and measures designed to establish and preserve database confidentiality, integrity, and availability. This article will focus primarily on confidentiality since it's the element that's compromised in most data breaches.

7.2 APPLICATION SECURITY

Application security is the process of making apps more secure by finding, fixing, and enhancing the security of apps. Much of this happens during the development phase, but it includes tools and methods to protect apps once they are deployed. This is becoming more important as hackers increasingly target applications with their attacks.

Application security is getting a lot of attention. Hundreds of tools are available to secure various elements of your applications portfolio, from locking down coding changes to assessing inadvertent coding threats, evaluating encryption options and auditing permissions and access rights. There are specialized tools for mobile apps, for network-based apps, and for firewalls designed especially for web applications.

7.3DATA SECURITY

Data security is the practice of protecting digital information from unauthorized access, corruption, or theft throughout its entire lifecycle. It’s a concept that encompasses every aspect of information security from the physical security of hardware and storage devices to administrative and access controls, as well as the logical security of software applications. It also includes organizational policies and procedures.

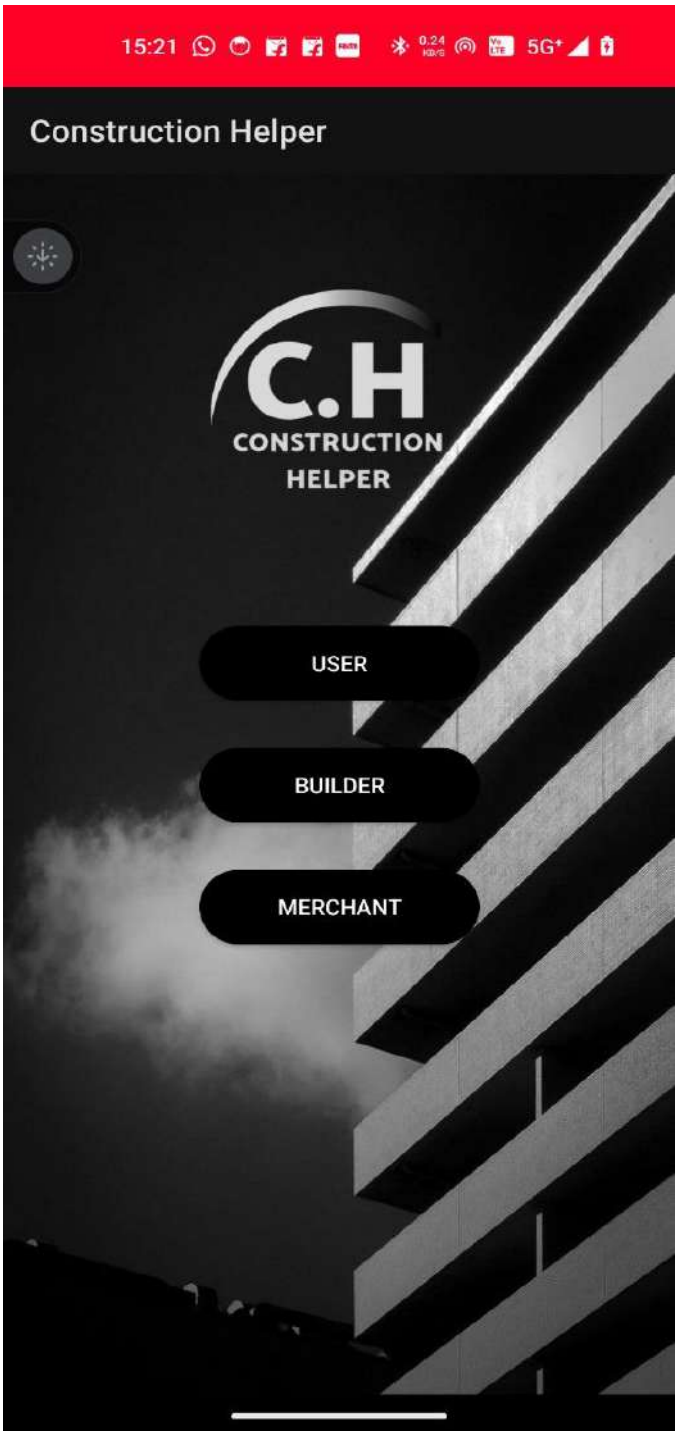
When properly implemented, robust data security strategies will protect an organization’s information assets against cybercriminal activities, but they also guard against insider threats and human error, which remains among the leading causes of data breaches today. Data security involves deploying tools and technologies that enhance the organization’s visibility into where its critical data resides and how it is used. Ideally, these tools should be able to apply protections like encryption, data masking, and redaction of sensitive files, and should automate reporting to streamline audits and adhering to regulatory requirements.

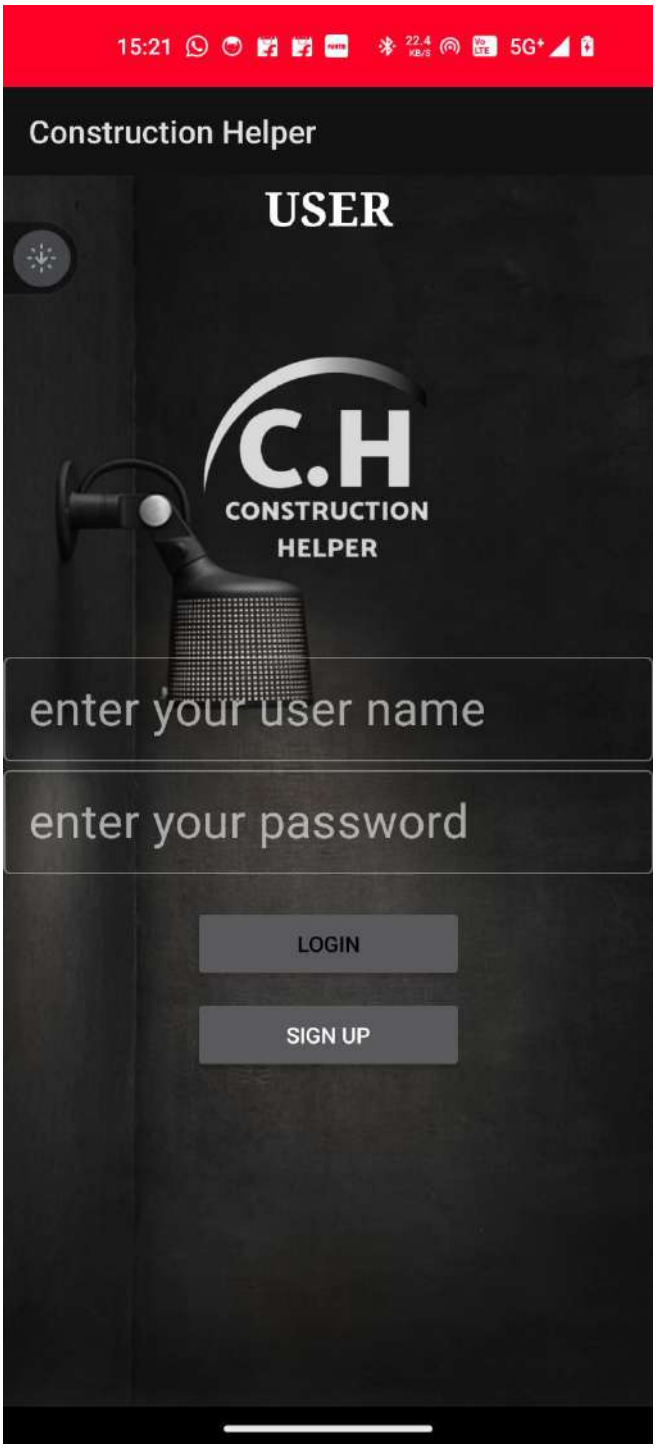
7.4 USER SECURITY

User Security consists of the platforms which protect your organisation's users, endpoints and their online activity to more efficiently correlate threats. As users are increasingly logging in to networks via their personal devices, securing these is just as important as securing company owned devices.

SCREEN LAYOUT

USER INTERFACE AND USER LOGIN PAGE



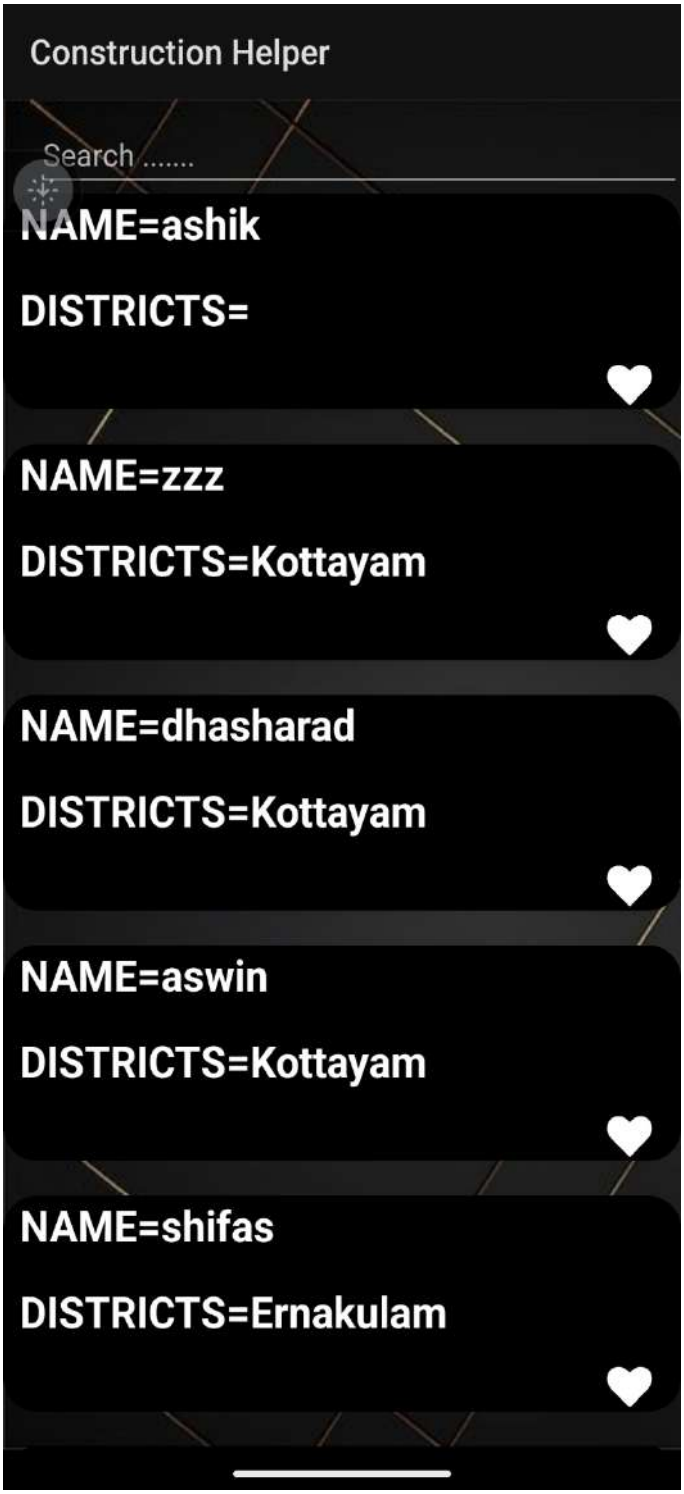


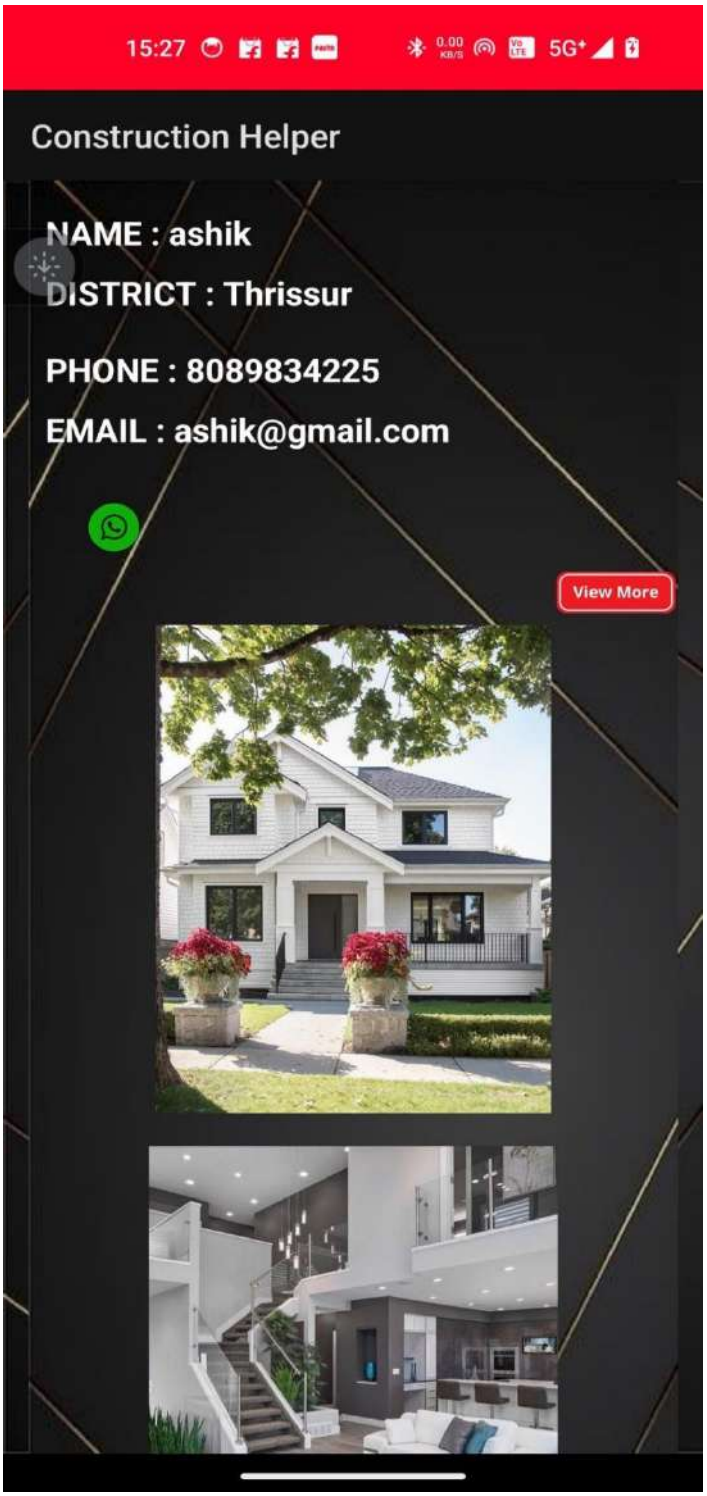
USER REGISTER PAGE /USER HOME PAGE



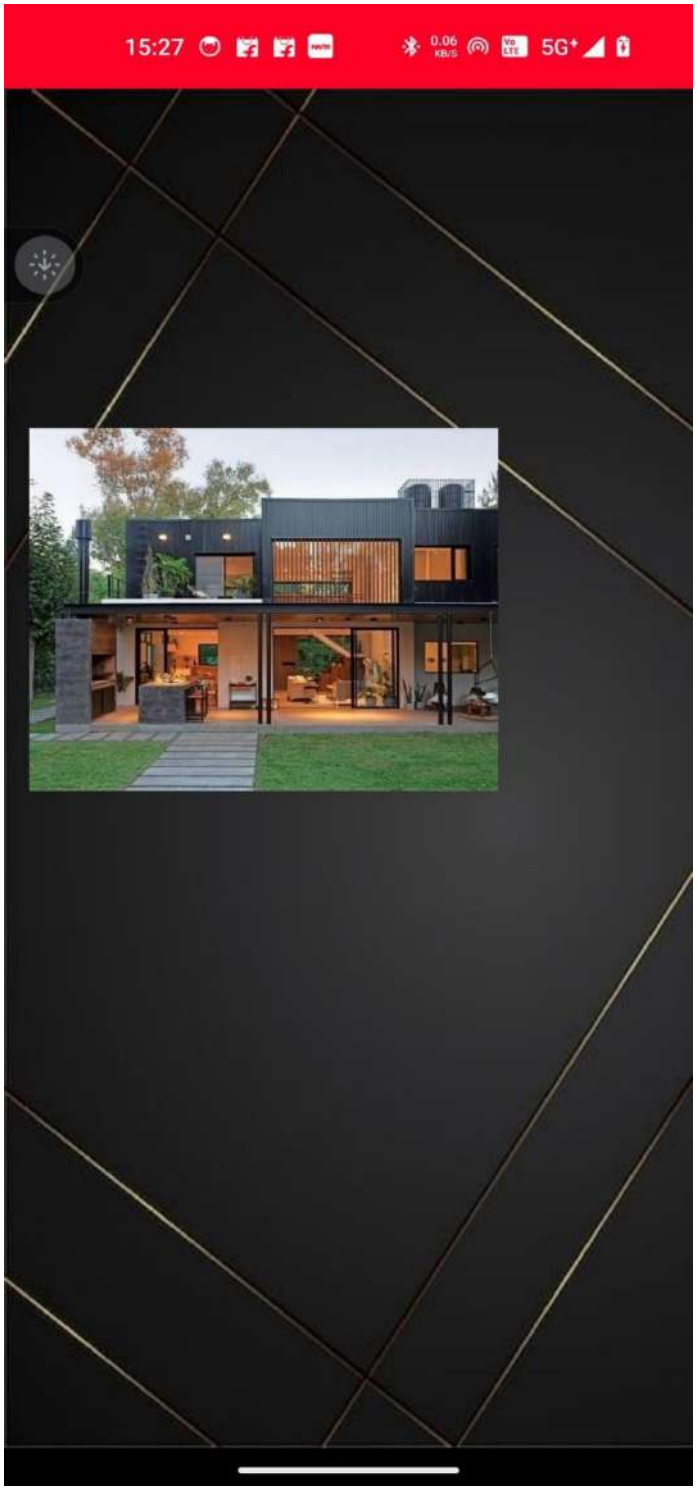


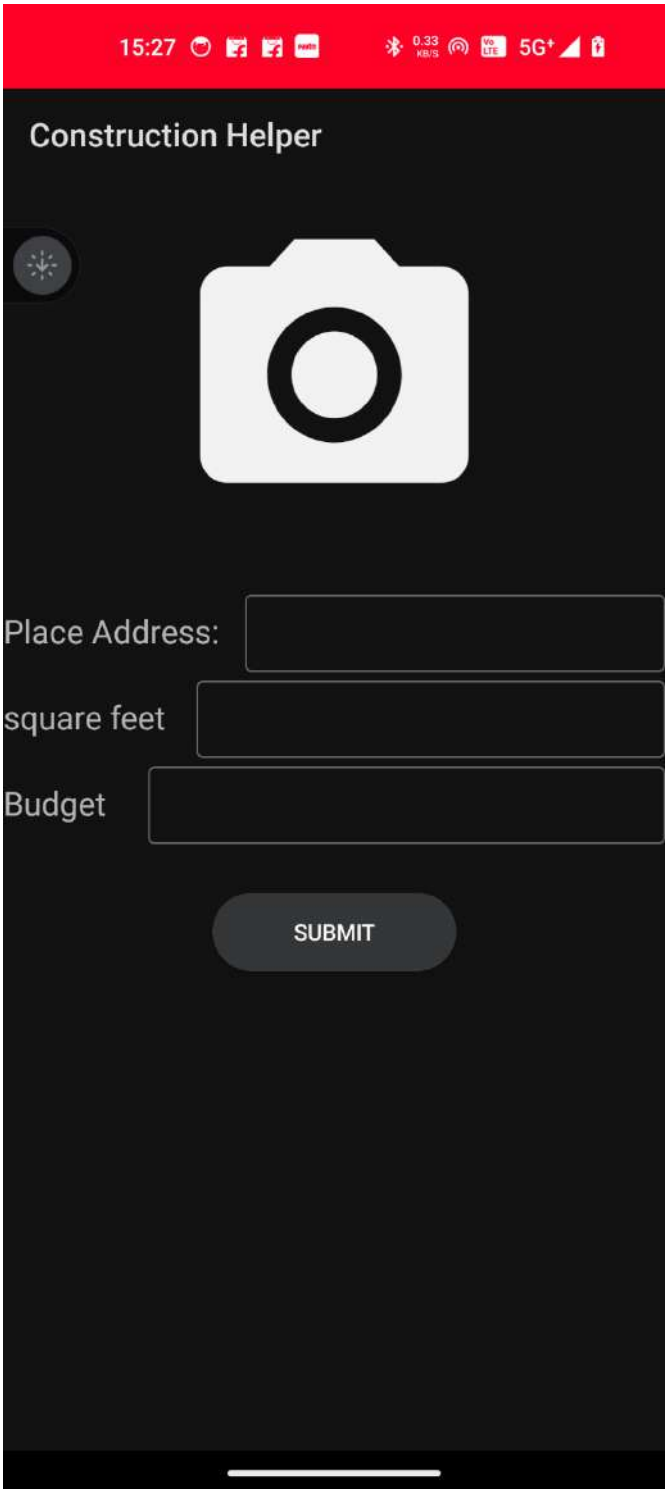
USER VIEW BUILDERS LIST/ BUILDER DETAILS AND VIEWWORKS



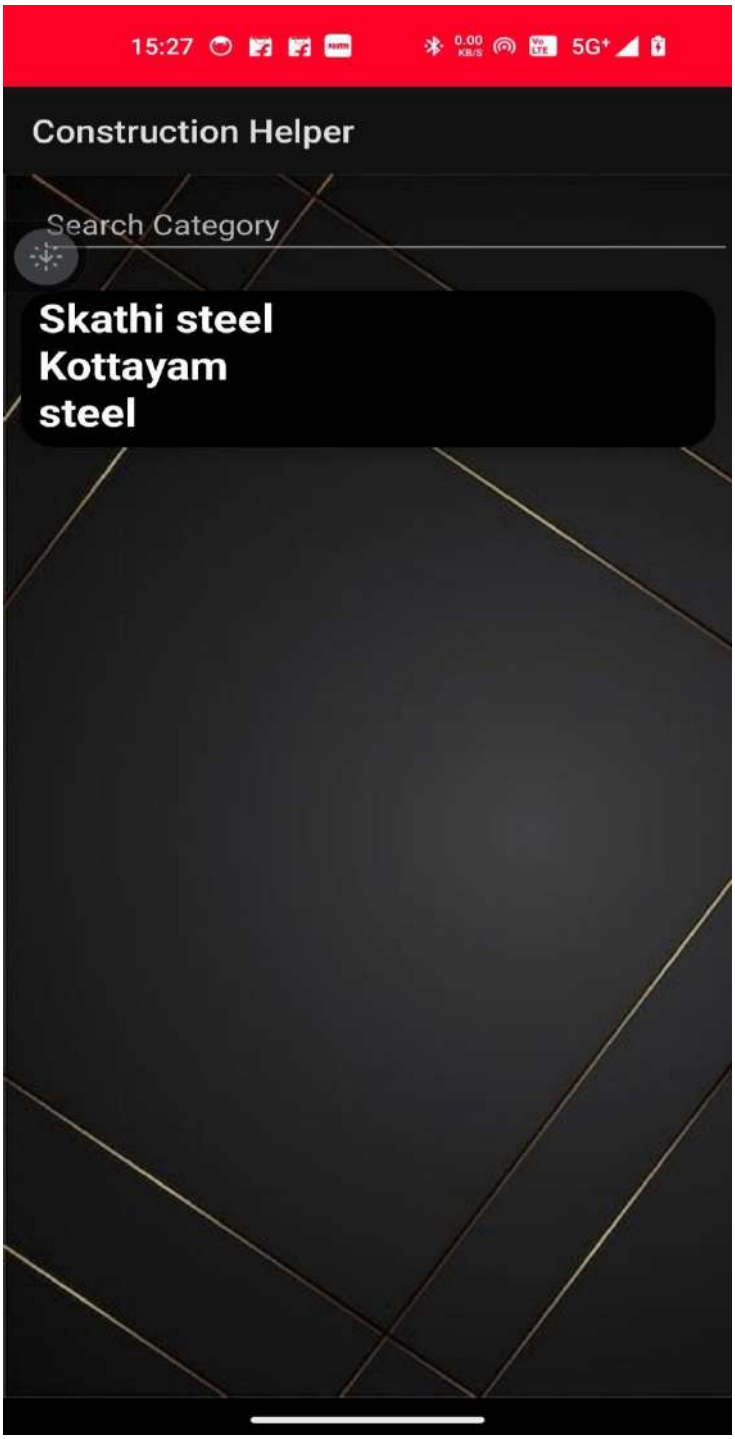


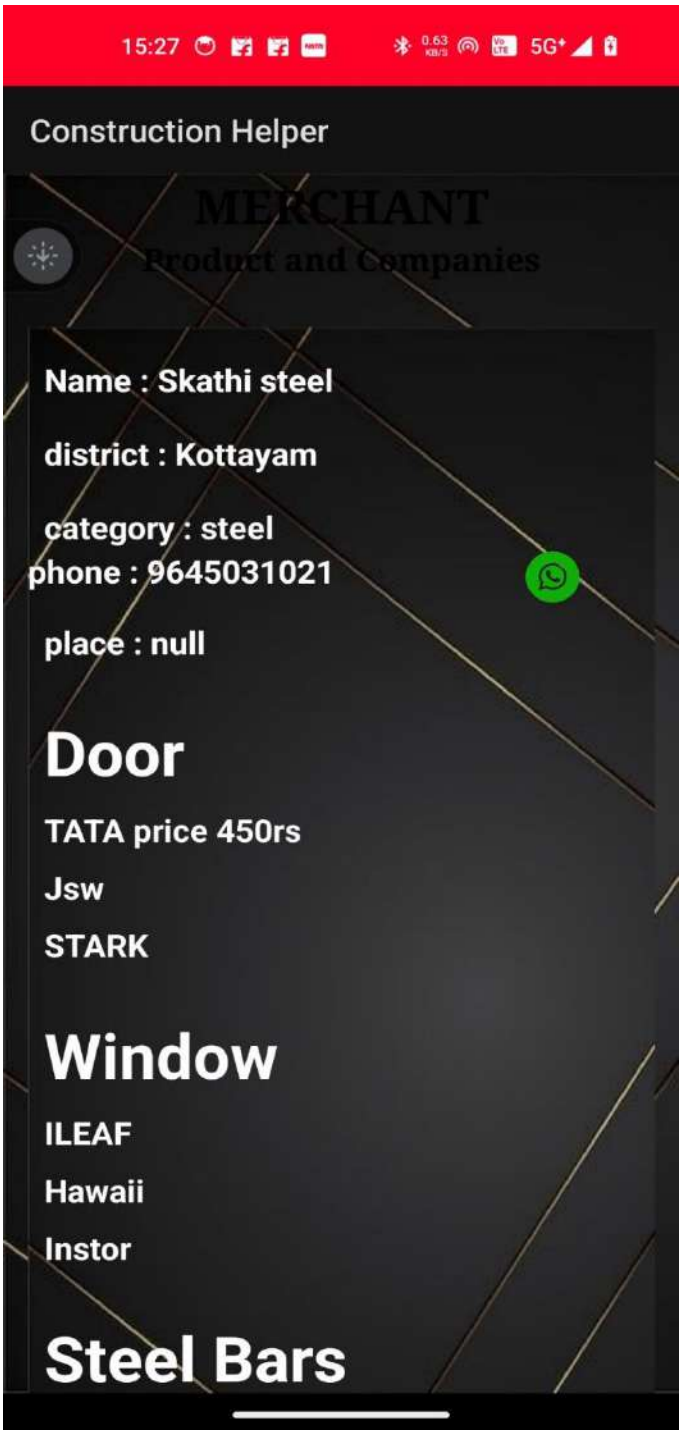
VIEW MORE IMAGES/ USER UPLOAD PLOT DETAILS





MERCHANT LIST/ MERCHANT DETAILS AND PRODUCT COMPANIES



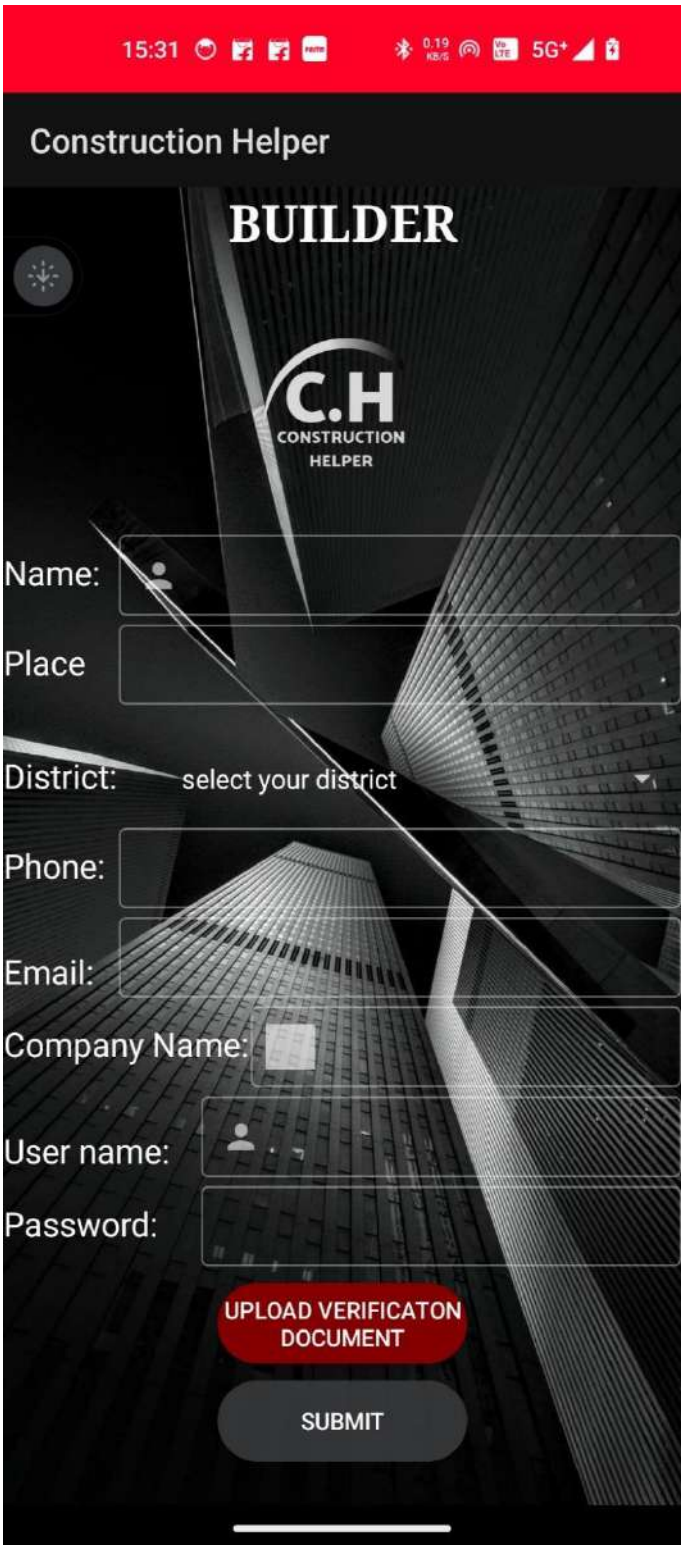


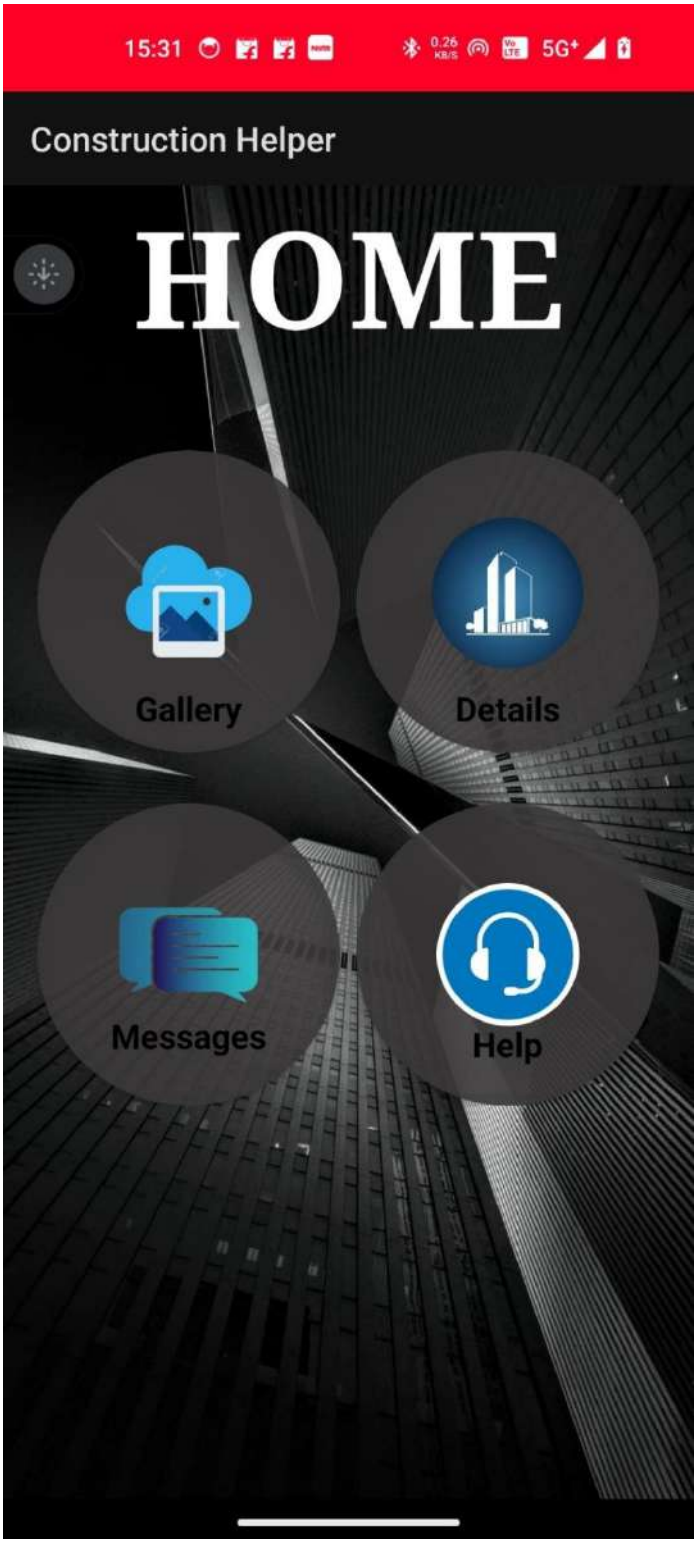
USER COMPLAINT/ BUILDER LOGIN PAGE





BUILDER REGISTER PAGE/ BUILDER HOME PAGE



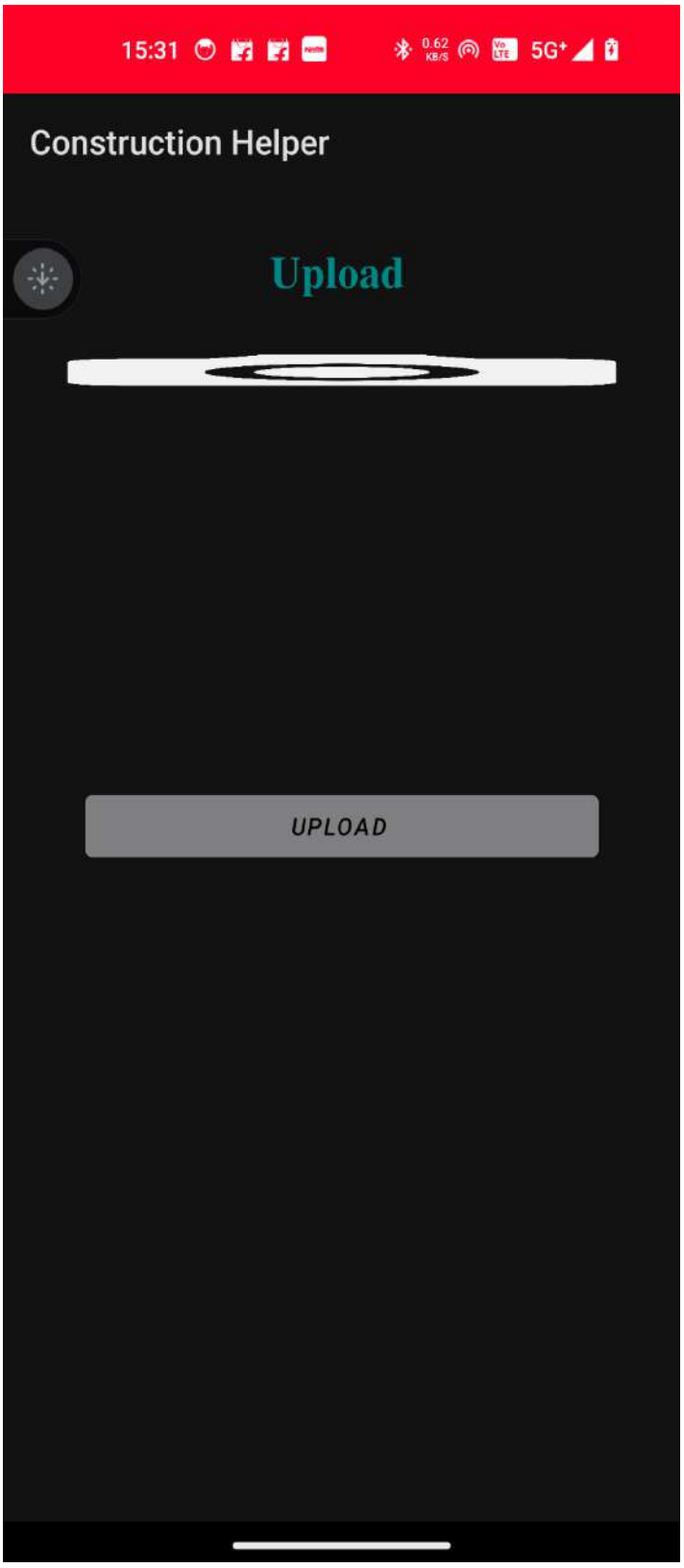


BUILDER VIEW WORKS IMAGES/ BUILDER UPLOAD WORKS





BUILDER UPLOAD MORE WORK IMAGES/ VIEW AND UPLOADED MORE IMAGES AND DELETE

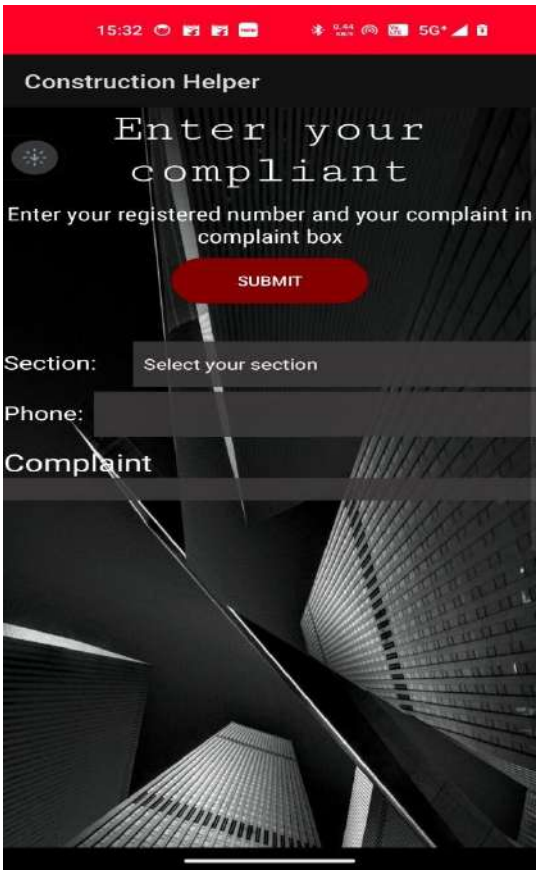




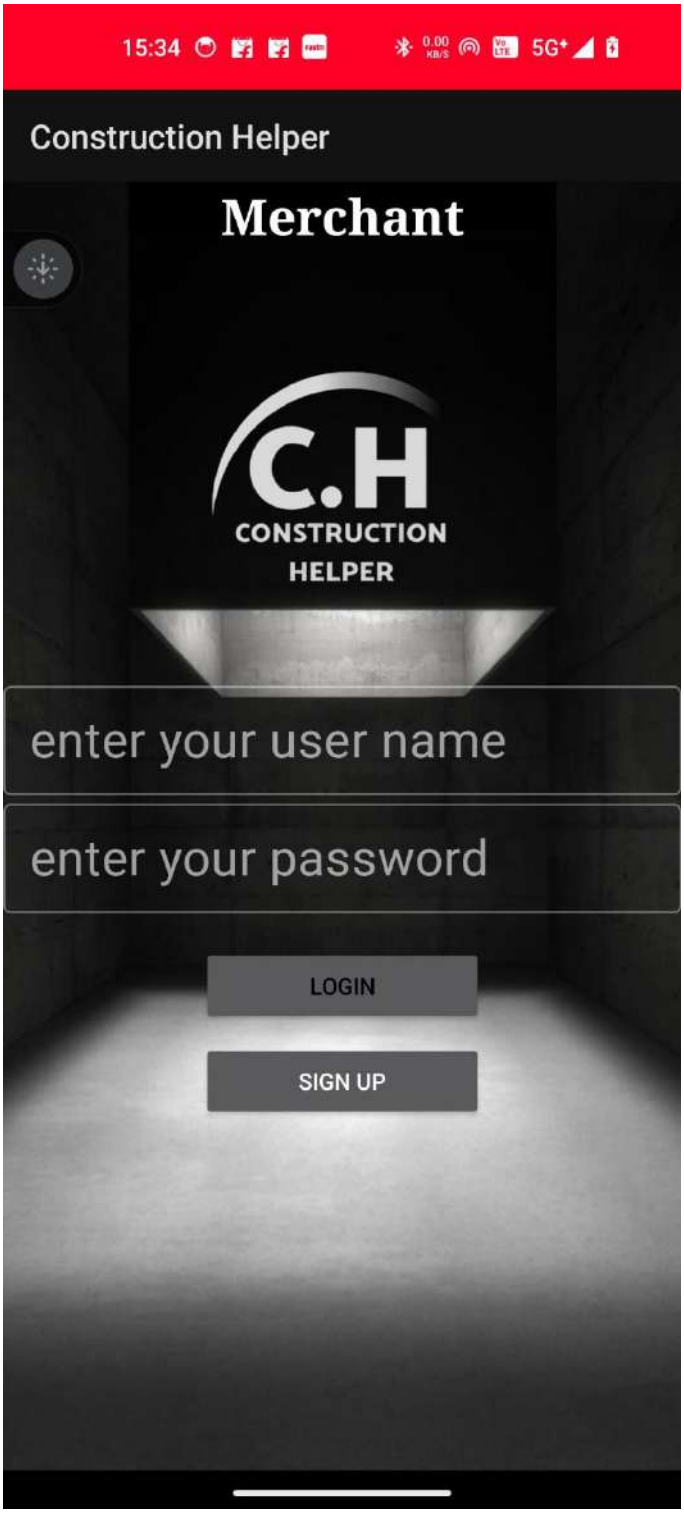
BUILDER DETAILS AND USER BOOKING/ BUILDER COMPLAINT





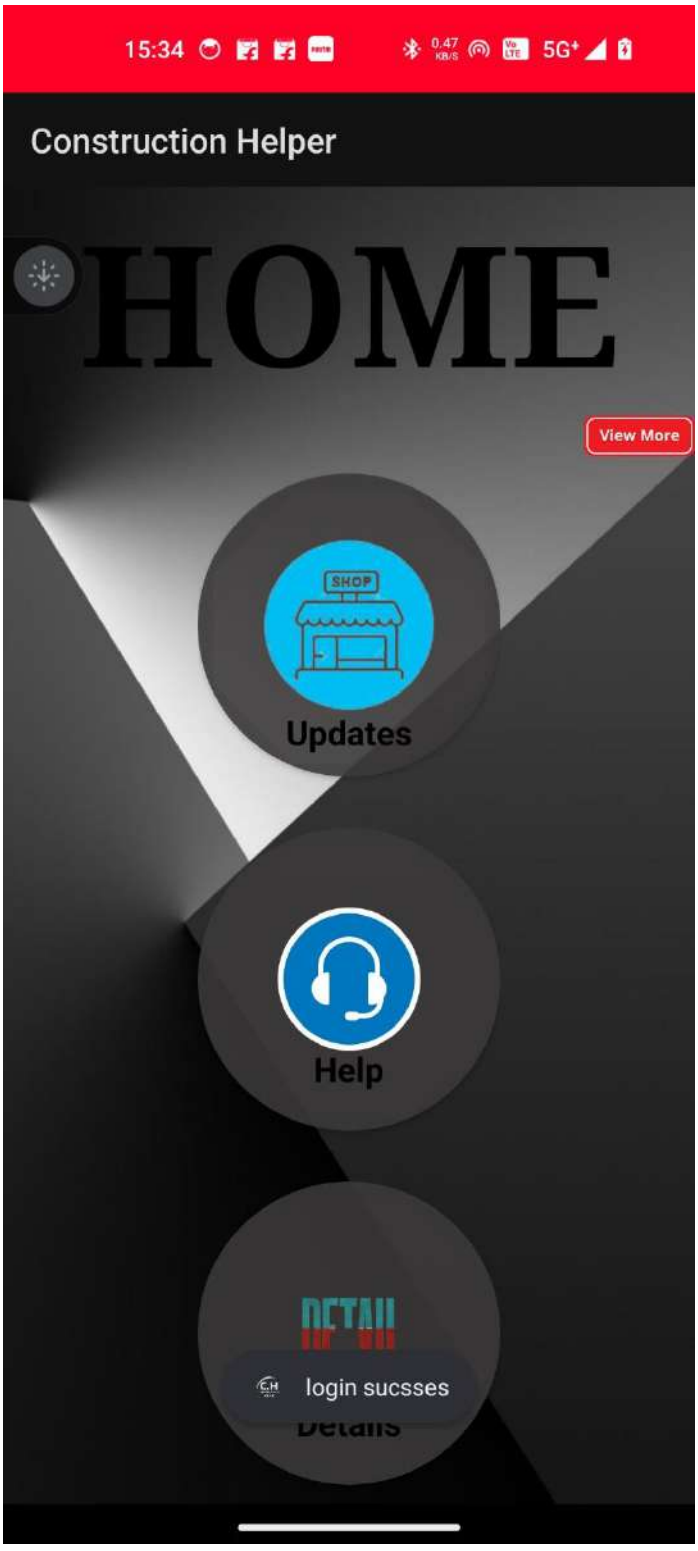


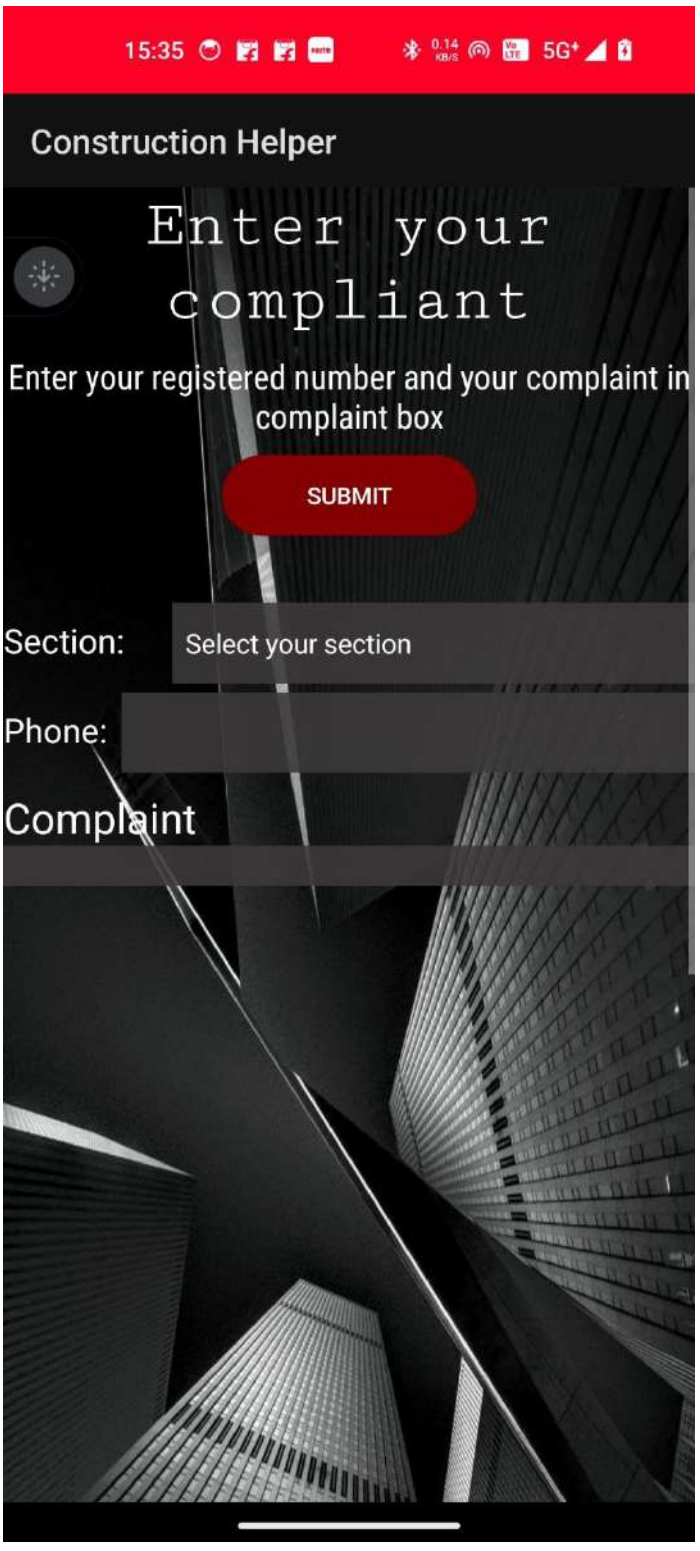
MERCHANT LOGIN AND REGISTRATION PAGE





MERCHANT HOME PAGE AND COMPLAINT REGISTER PAGE





MERCHANT UPLOAD PRODUCTS, COMPANY AND VIEW



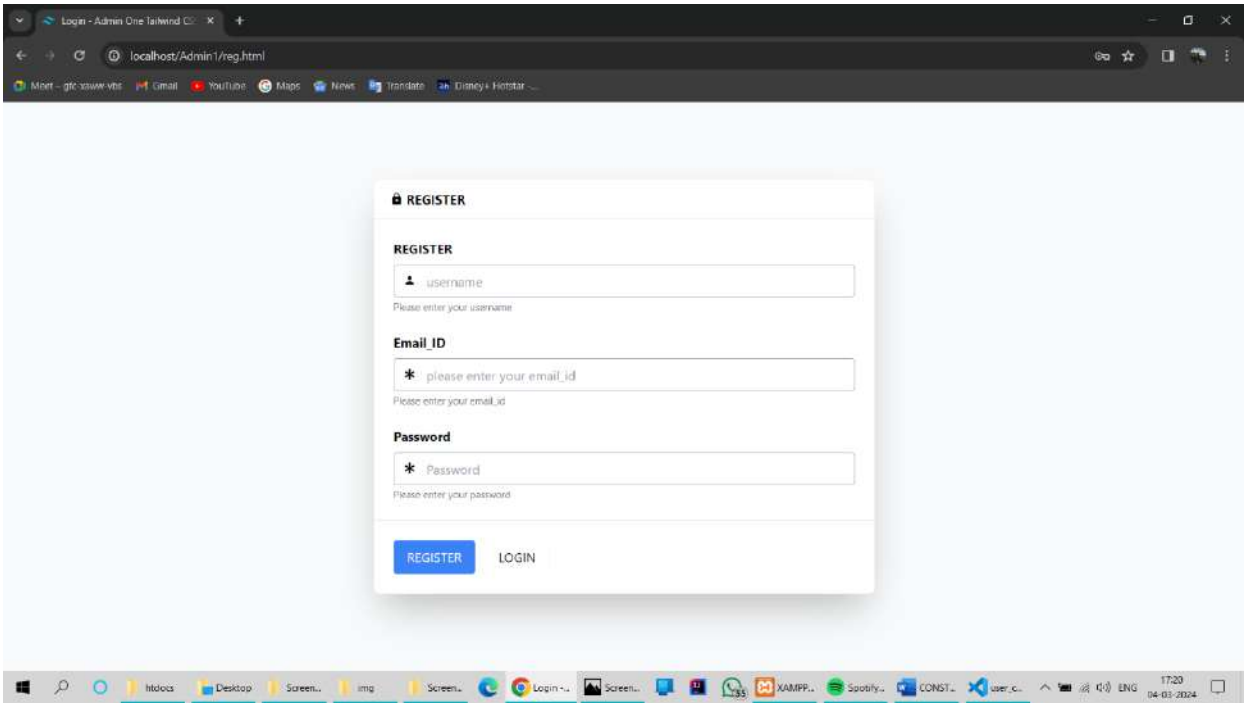




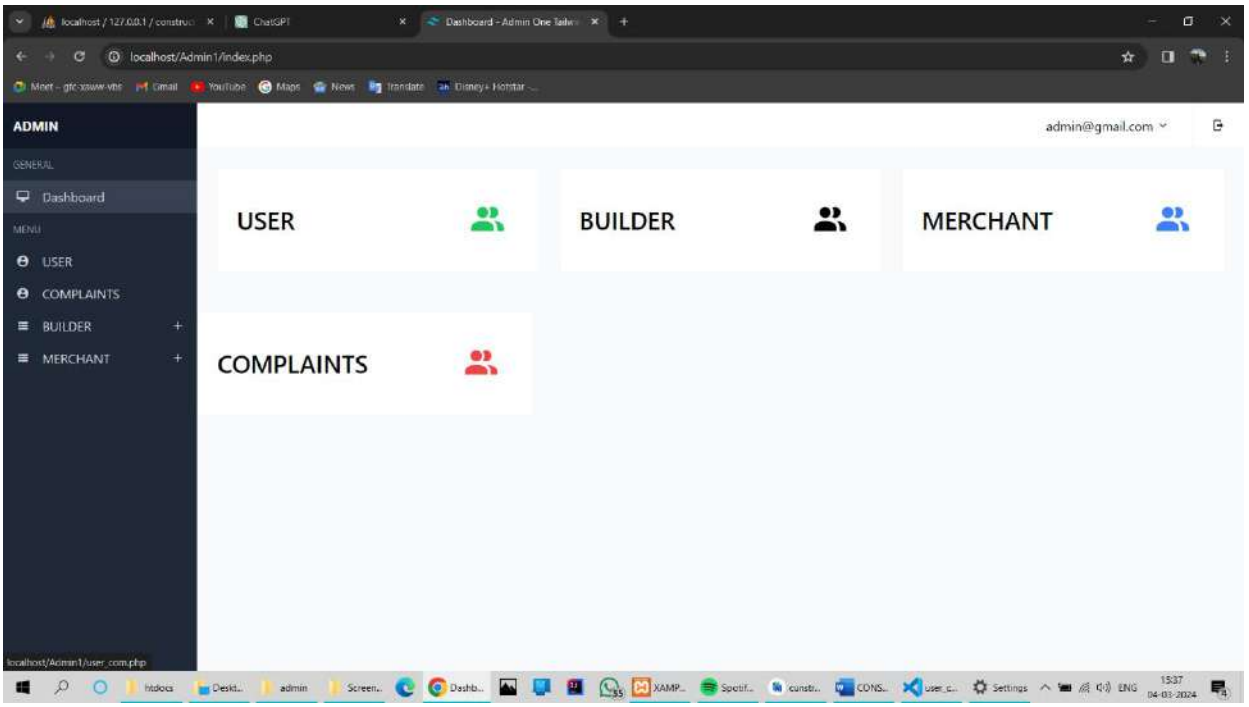


MERCHANT DETAILS UPDATE/ ADMIN REGISTER

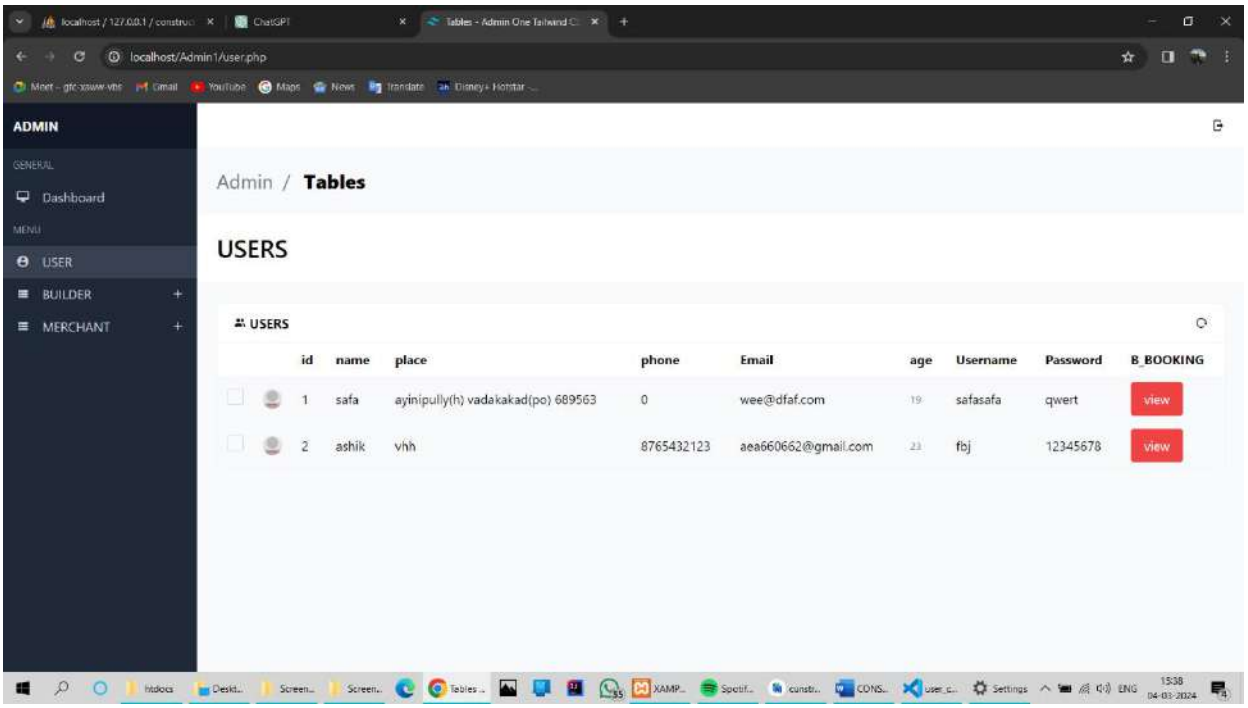




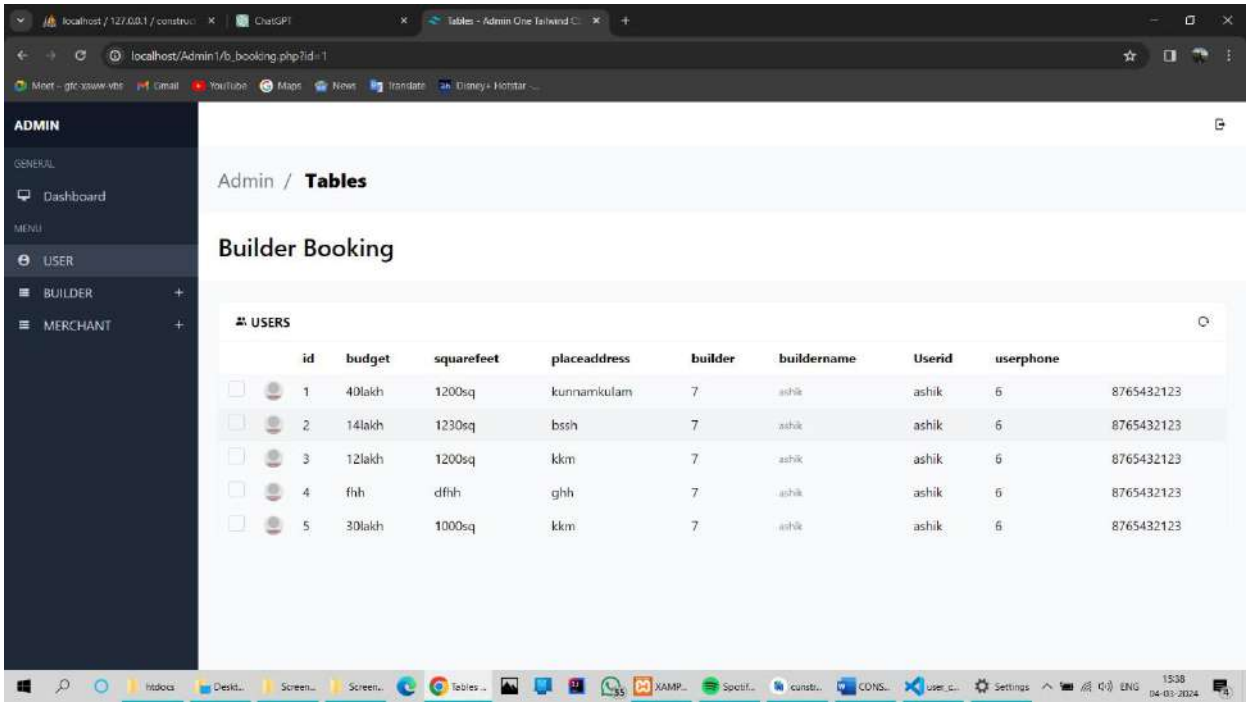
ADMIN PANEL



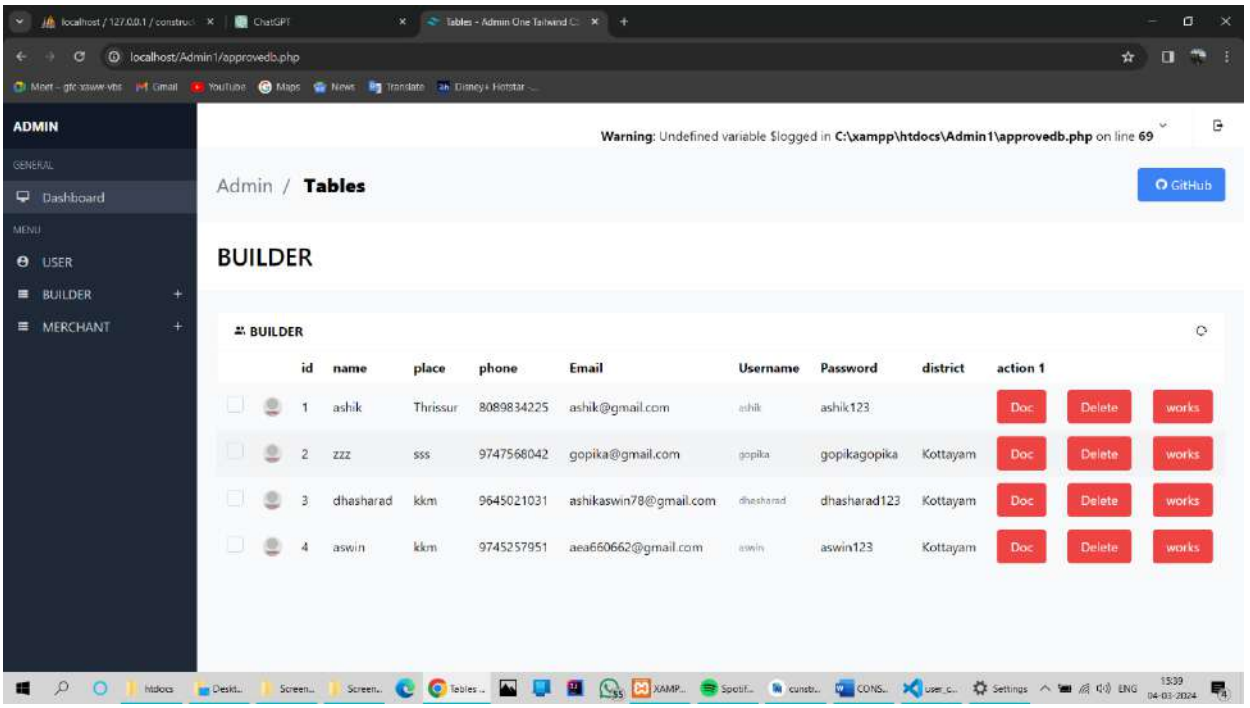
USERS LISTED



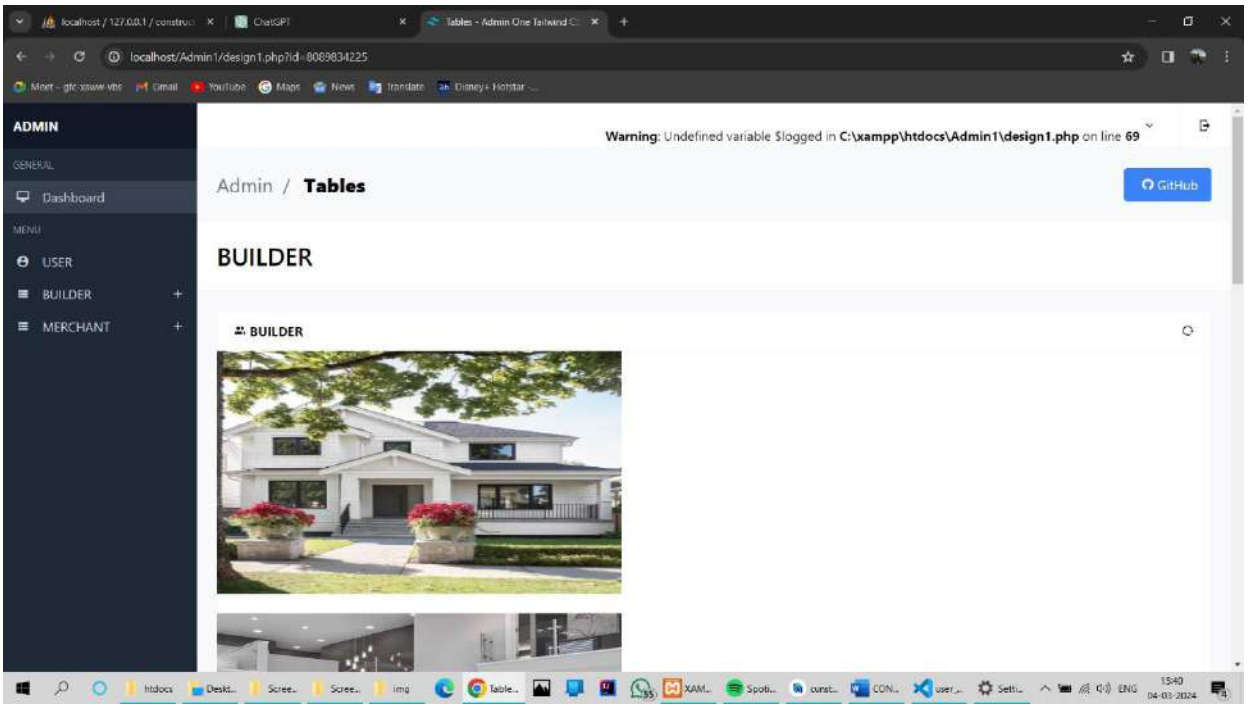
USER UPLOADED PLOT DETAILS VIEW IN ADMIN

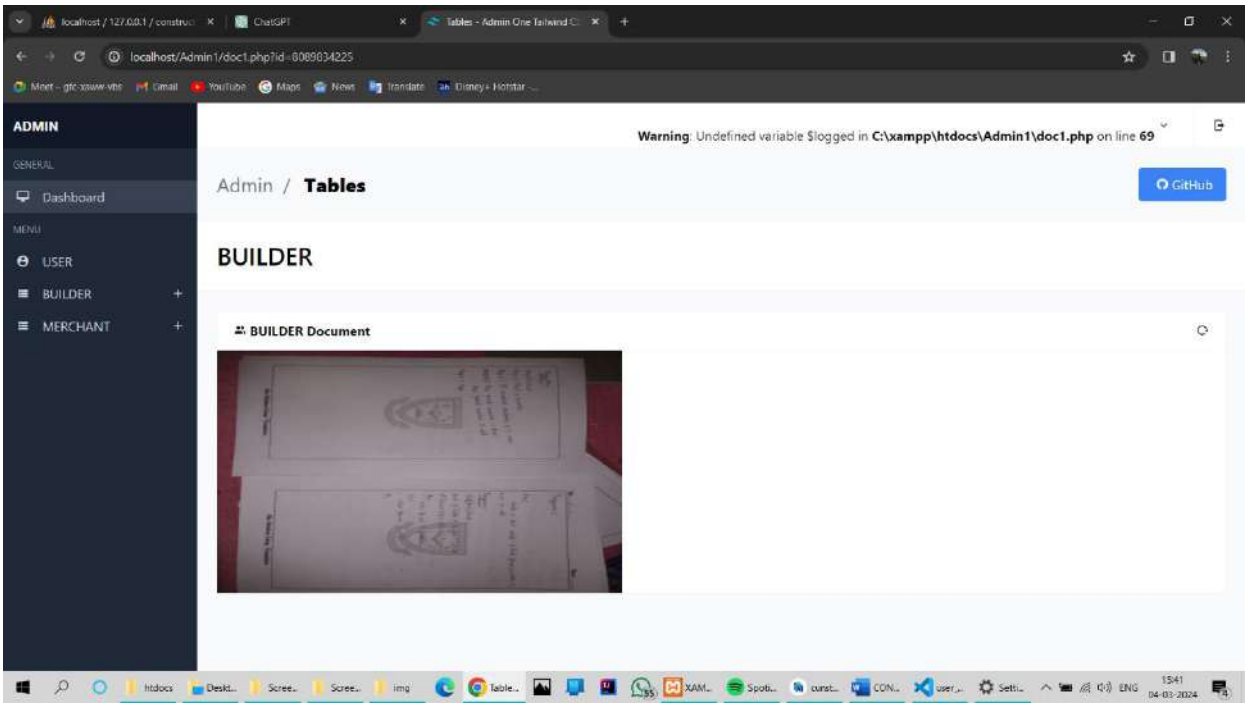


BUILDER DETAILS



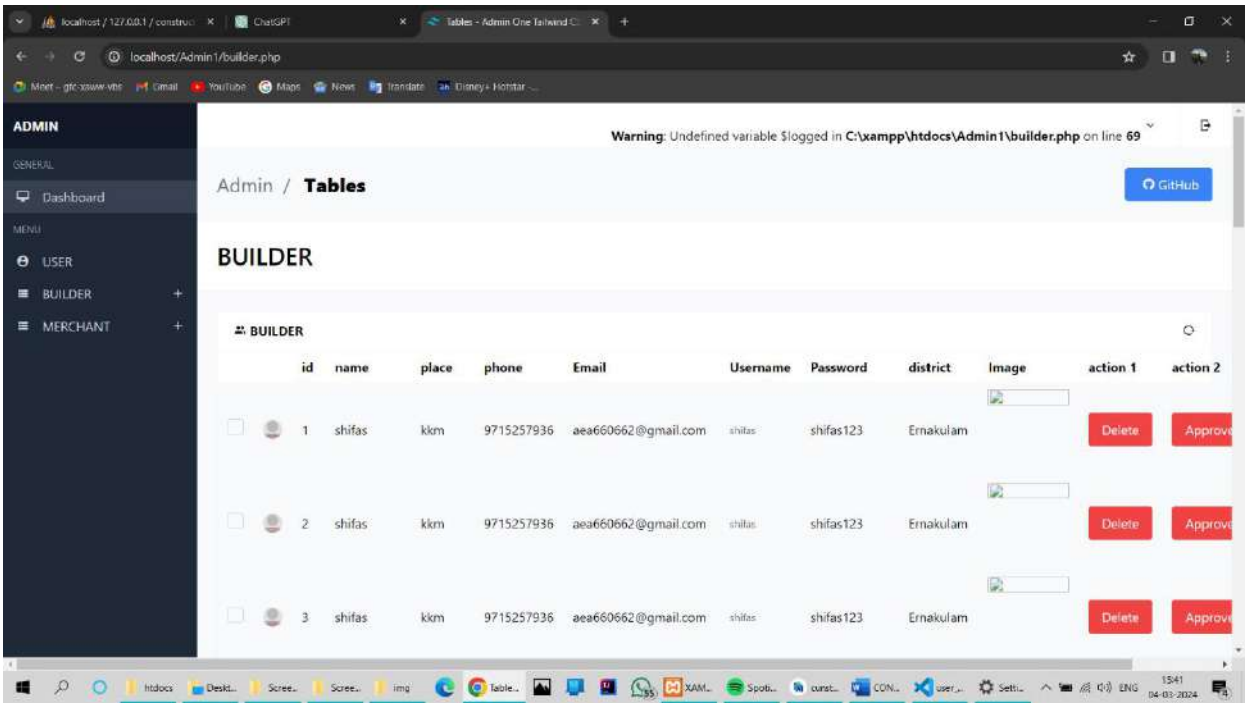
BUILDER IMAGES ARE VIEW ADMIN



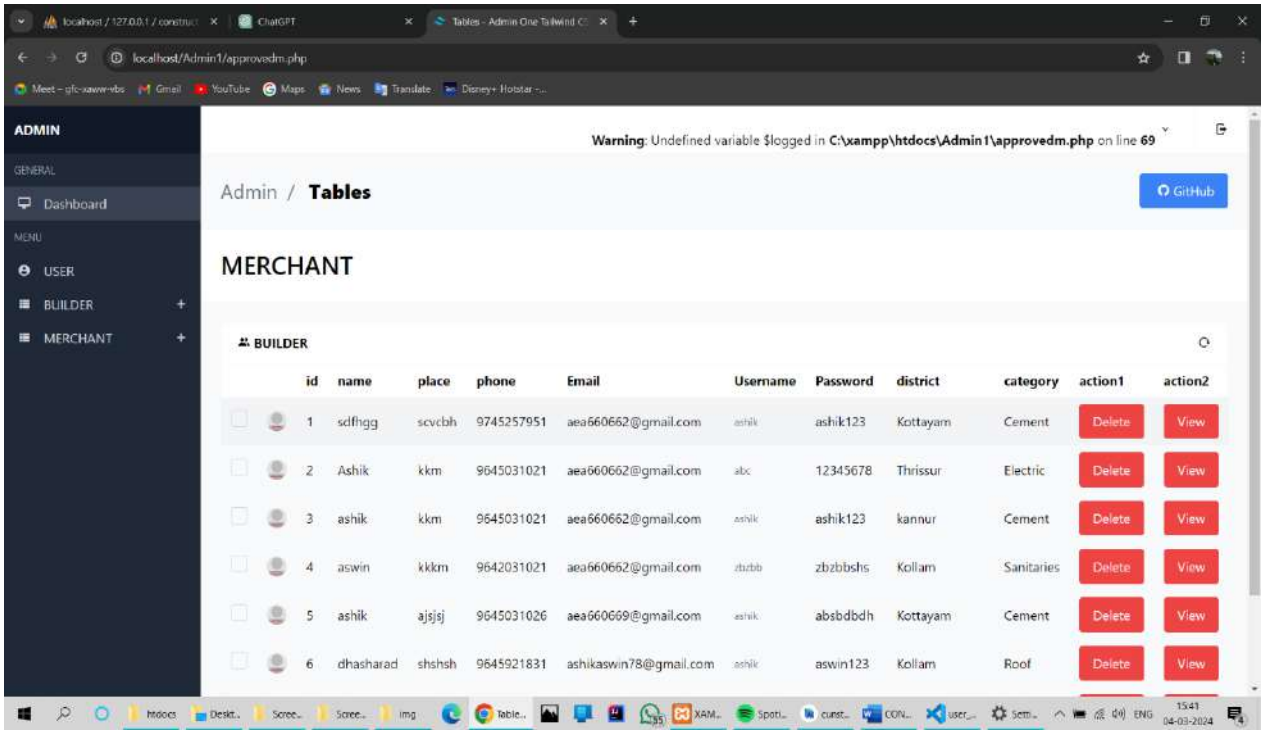


BUILDER DOCUMENT VERIFICATION

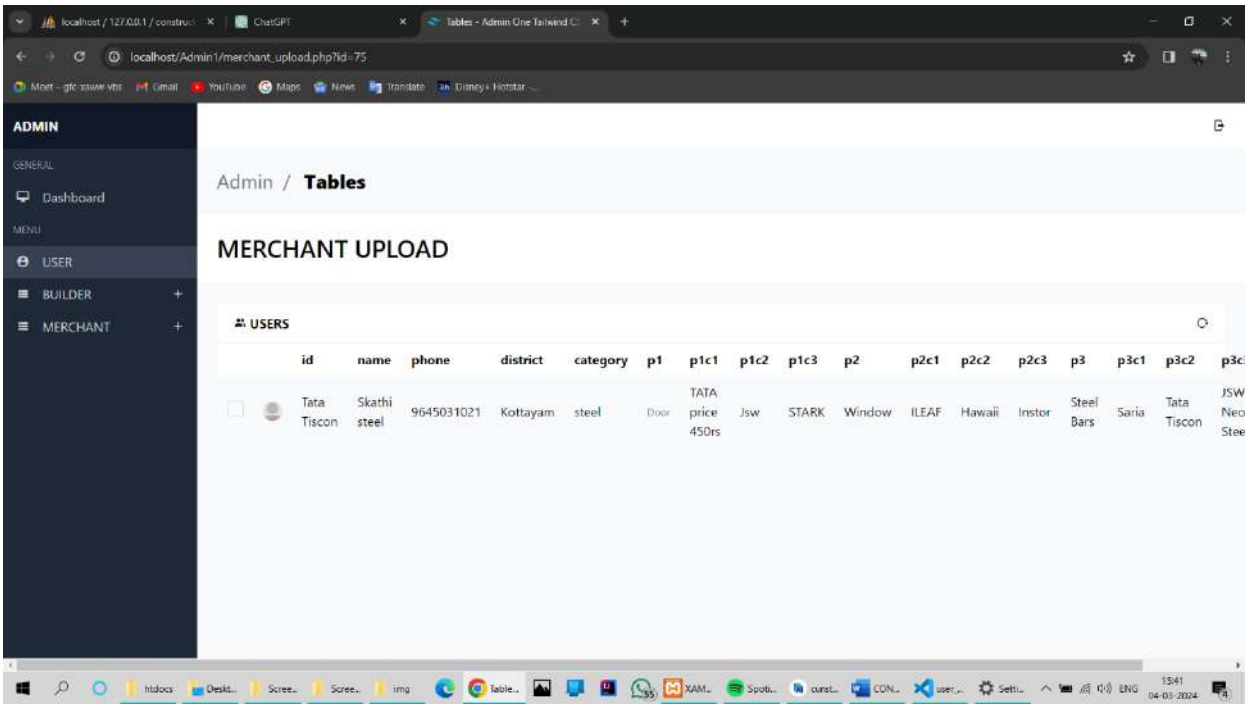
BUILDER APPROVE BY VERIFING DETAILS



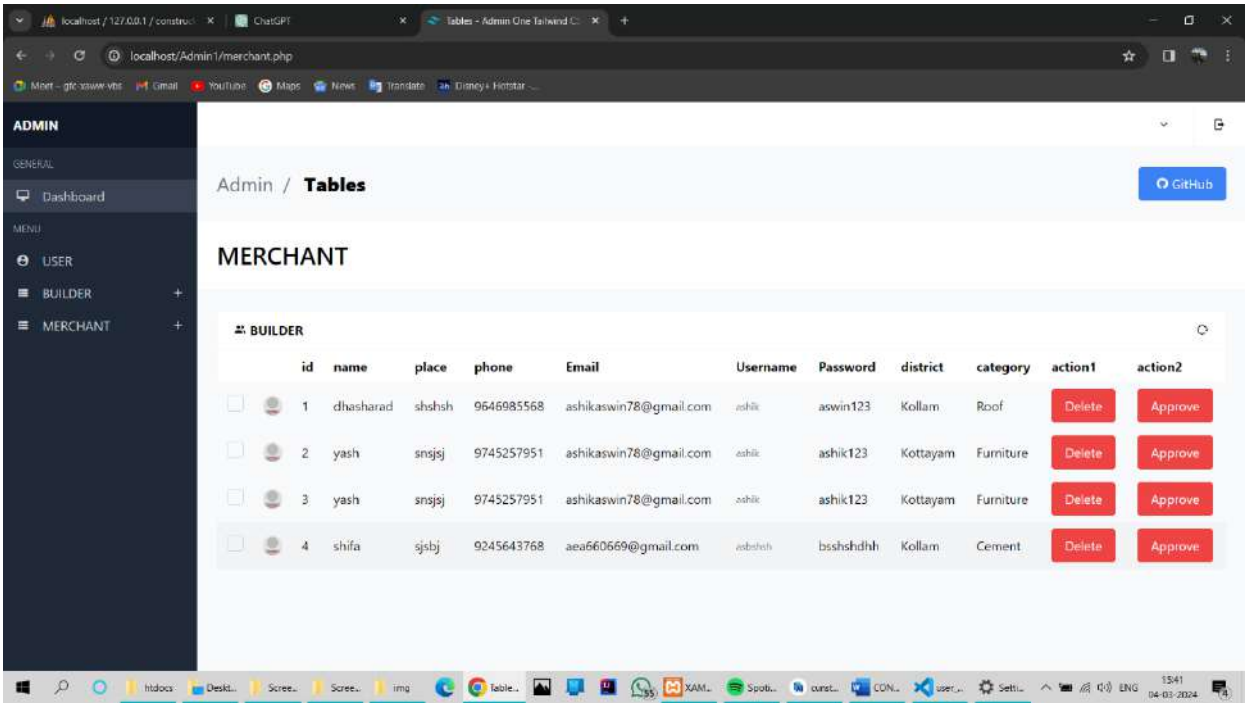
MERCHANT DETAILS



MERCHANT PRODUCT DETAILS



MERCHANTS ARE VRIFING AND APPROVE



CONCLUSION

9. CONCLUSION

The project Construction helper System was completed with the timespan.It is developed using Android technology fully meets the objectives of the system which it has been developed.Construction helper can be a valuable thing for connecting real people with Builder and Merchants ,especially for Common people , who wants to Built their Buildings or Houses with the help of Builders and Merchants.The main purpose of Construction Helper is to provide the facility to build for getting the Builders and Merchants.so it enables applicants to search the Builders and Merchants on a convenient manner and to enable User to find the suitable Builder and Merchants .

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10. BIBLIOGRAPHY

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